MedAustron

Non-Clinical Research Strategy

2022 – 2023 – 2024

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Preface

The second period of Non-Clinical Research at MedAustron 2019 – 2021 produced high quality experimental and innovative research with increasing visibility and recognition in the international community. This is well documented in the first Review part of the new strategic plan for non-clinical research 2022 – 2024. The multidisciplinary research strategy integrating various disciplines from physics, biology, computer and medical science for well defined projects proved to be successful under the leadership of scientific experts from these disciplines from key scientific Austrian institutions. It has been essential that these efforts could be further embedded into long standing research expertise from major players in international research such as the Medical University of Vienna, the Technische Universität (TU) Wien, the Institute of High Energy Physics and the Medical University of Graz. Such a multi-disciplinary and multi-institutional research strategy seems therefore to be key for a continuous growth of non-clinical research at MedAustron into a "healthy" future research landscape in Austria and beyond.

During the second research period from 2019 until 2021, new functionalities were made available for non-clinical research, i.e. a fully commissioned carbon-ion beam with clinical beam parameters and the 800 MeV proton beam. In addition, reduced proton fluxes in comparison to those in clinical use were established. New professorships for Accelerator Physics and Molecular Radiation Biology were implemented at the TU Wien and the Medical University of Vienna. Many young scientists have to be acknowledged who successfully contributed essentially at different levels (postdoc, PhD, Master, Bachelor). The contribution and commitment of the University of Applied Sciences Wiener Neustadt has also become crucial in the context of small animal imaging. The link between non-clinical research activities and MedAustron was intensified by embedding research and development activities motivated by current clinical needs.

For the upcoming three years, the third "MedAustron Non-Clinical Research Strategy 2022 – 2024" describes the strategy and the intended research projects based on the experience of the past six years of non-clinical research activities at MedAustron, which provides a sound basis for a successful future development plan. Most research projects from the last research period will be continued and extended. Several of these projects received third party funding, which is key for further growth. They are now organised in four pillars: Applied Particle and Medical Physics; Biophysics and Molecular Radiobiology; Technological Innovations and Clinical Implementation; Accelerator Physics. This more focused organisation facilitates the overall management and also strengthens the national and international visibility of non-clinical research at MedAustron. Several of these projects can be expected to become "flagship projects" in the international community (e.g. the development of new technology for ocular treatment). There is also room for new approaches and extensions such as the important launch of in vivo experiments at MedAustron in the next years. Additionally, the performance of the particle accelerator will be continuously increased providing new features for the users. One example is the commissioning of helium-ion beams, which enlarges, in a first step, the opportunities for non-clinical research considerably and will bring MedAustron in the forefront of ion-beam research worldwide.

For the coming period the availability of more beam time and the enriched functionality needs to be supported by an increase in human resources. We are grateful for the contributions of the big research partners involved in the multi-institutional MedAustron research for their input so far and in future. We are confident in the funding activities of the established research

groups at MedAustron so far and in future. There is still a dream alive from the early days of non-clinical research at MedAustron to receive substantial long-term third party funding for this outstanding national research project in Austria.

Concerning the organisation of research, the contribution of Thomas Schreiner has been outstanding and will be crucial. For the overall scientific coordination, the role of Dietmar Georg has been substantial and will be key for the advancement of non-clinical research at MedAustron.

For the Advisory Board, I am grateful for the continuous active support of Gerald Badurek from the TU Wien and the support of Dr. Michaela Fritz, Vice Rector of Research at the Medical University of Vienna and Prof. Johannes Fröhlich, Vice Rector for Research at the TU Wien, who joint the Advisory Board during the last research period. In addition, I would like to mention several national and international research experts who supported us in evaluating and advising most research projects carried out during the last research period.

Richard Pötter
Chairman
Advisory Board MedAustron Research

Authors and Contributors

Contributions from all coordinators and work-package leaders, as listed in the respective sections, as well as contributions from researchers of the following organisations are greatly acknowledged:

- Medical University of Vienna
- TU Wien
- Institute of High Energy Physics, Austrian Academy of Sciences:
- University of Applied Sciences Wiener Neustadt
- Medical University of Graz
- EBG MedAustron

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Introduction

Review and Vision

1 Review Research Strategy 2019 – 2021

1.0 Commissioning and Quality Assurance Procedures

Equipment Commissioning for Carbon Ions

Similar to the commissioning of the dosimetric equipment for protons in 2016, phantoms and detectors were tested and calibrated for the respective use with carbon ions in the clinically used energy range from 120 MeV per nucleon to 402.8 MeV per nucleon. In summer 2019, several acceptance measurements were carried out in order to compile a comprehensive report documenting the availability of carbon ions in irradiation room 1 (IR 1). The verified parameters were the same as already performed for protons in 2016[1]. After finishing, the acceptance report for carbon ions[2] was forwarded to the representatives of the Austrian Federal Ministry of Education, Science and Research in November 2019. Carbon ion basic beam data for the beam model were acquired from November 2019 until January 2020 and the beam model created by RaySearch (RaySearch Laboratories, Stockholm, Sweden) was validated until May 2020. During the commissioning also students were involved, working on their BSc, MSc and even PhD theses. After the successful commissioning of the RayStation treatment planning system (TPS), further investigations with complex heterogeneous phantom geometries in the TPS were performed. The first studies with protons in 2019 and 2020 [2] were then continued with carbon ions within the scope of a PhD work from Sirinya Ruangchan in 2020 and 2021.

High-Energy Proton Commissioning

Besides the availability of carbon ions in IR 1, the maximum proton energy of 800 MeV was proven with a dedicated measurement set-up using a 190 cm long plastic absorber and the Peakfinder for determining the residual penetration depth in water. The respective results and the comparison with simulation data are documented in the acceptance report[2] mentioned above. In addition to the nominal intensity of 10^{10} protons per spill (100% transmission of the beam, i.e. no degrader), the required low flux (10^4 protons per second) could be commissioned besides the clinical energies also for 800 MeV protons. A dedicated detector set-up

[1] See "Acceptance Report for Protons in IR 1, PM-160927-TSC", December 2016.
[2] See "Acceptance Report for Carbon Ions in IR 1, PM-190920-TSC", November 2019.

and an analysis tool were established in order to perform regular quality assurance (QA) measurements for these low flux settings.

Treatment Planning and Imaging for Small-Animal Irradiation

A dedicated small animal TPS for X-ray irradiation (i.e. µ-RayStation) was purchased in 2021. The commissioning of this TPS started afterwards with studies investigating the effect of different collimator sizes and beam incidence directions on the dose distribution. This way, a synergy between treatment planning, dose delivery, and workflow creation is envisaged so the ideal beam configuration for the X-ray irradiator can be found. A first version of this set-up was designed and is currently improved.

Investigations for dose delivery to small targets with protons were initiated in 2020 – 2021. Collimators with different openings, which shield the healthy tissue, were placed in front of the target. Target sizes in the range from 2 cm down to 0.2 cm were investigated on a simplified phantom, consisting of two different tissue equivalent materials. The preliminary results indicated sufficient dose coverage for the targets as small as 0.5 cm. This target size is in line with the typical treatment volume relevant for radiobiological pre-clinical in vivo studies.

A modular bench-top pre-clinical imaging system, comprising a micro-positron emission tomography (PET), micro-single-photon emission computed tomography (SPECT) and micro-computed tomography (CT) (Molecubes, Ghent, Belgium) was purchased, installed and integrated in the research infrastructure at MedAustron in June 2019 by the University of Applied Sciences Wiener Neustadt. Technical implementation, calibration and specific system commissioning of the micro-CT imaging modality started in September 2019. The development and verification of suitable micro-CT acquisition- and reconstruction protocols for small-animal irradiation treatment planning were performed with technical phantoms, as well as first in vivo measurements. The system commissioning and calibration of the micro-SPECT and micro-PET imaging modality, the development of a suitable experimental workflow and the establishment of appropriate image registration for intermodal image fusion were performed in June 2020 at the research laboratories of the University of Applied Sciences Wiener Neustadt and the Pre-Clinical Imaging Laboratory of the Medical University of Vienna. A thorough system performance analysis of the micro-CT, micro-PET and micro-SPECT imaging modalities according to the National Electrical Manufactures Association (NEMA) NU 4-2008 guidelines started in January 2021. A central issue for the dedicated imaging-based small-animal irradiation workflow is the establishment of an appropriate technical and experimental set-up for animal immobilisation. The development of a suitable workflow for animal immobilisation and anaesthesia was initiated in March 2021.

In January 2021, the relocation process of the 15.2 T ultra high-field (UHF) micro-magnetic resonance tomography (MRT) system (BRUKER, Ettlingen, Germany) from the Vienna Bio Centre Core Facilities to MedAustron was successfully finished. The subsequent technical integration and implementation, as well as the system calibration of the micro-MRT system started in April 2021. Technical problems with the machine delayed the technical implementation, but since October 2021, the MRT is functional. However, the required MRT coils are currently fixed by the vendor.

Establishing Specific Quality Assurance Procedures

QA procedures for clinically used beam parameters were performed in regular intervals to ensure the proper characteristics of proton and carbon-ion beams in IR 1. Along with the beam QA also equipment QA, either regular checks or tests after repairs, were carried out when required. Due to continued improvements on the accelerator side, e.g. increase of beam delivery efficiency, further acceptance and validation of the beam data was necessary after each update. Although publications based on QA data are not envisaged, beam time for performing QA measurements is important for the proper operation of the IR 1 beam line.

Monte Carlo Simulations of Carbon-Ion Beam Line

Monte Carlo simulations are especially useful for research purposes, allowing to help in experimental planning and consequently reducing the beam time required as well as to investigate effects not easily accessible using experimental measurements. A prerequisite is the availability of high-quality beam models, which may need to be adapted depending on the experimental focus. In order to efficiently perform beam modelling for Monte Carlo simulations, a system was developed, which was then successfully applied to the generation of a validated carbon-ion beam model for the research irradiation room at MedAustron [1]. Furthermore, this method can now be used to create custom beam models fine tuned to the experiment's requirements allowing maximum flexibility with limited modelling time.

Scientific Output

3 Publications

[1] H Fuchs, A Elia, AF Resch, et al. "Computer-assisted beam modeling for particle therapy". In: *Medical Physics* 48.2 (Feb. 2021), pages 841–851.

[2] S Ruangchan, B Knäusl, H Fuchs, D Georg, and M Clausen. "Experimental benchmarking of RayStation proton dose calculation algorithms inside and outside the target region in heterogeneous phantom geometries". In: *Physica Medica* 76 (2020), pages 182–193.

[3] S Ruangchan, H Palmans, B Knäusl, D Georg, and M Clausen. "Dose calculation accuracy in particle therapy: comparing carbon ions with protons". In: *Medical Physics* (2021), Accepted Author Manuscript.

10 Finalised Theses: 8 BSc, 2 MSc, 0 PhD

[4] M Aichberger. "Detector Cross Calibration in Carbon Ions". Bachelor's thesis. Radiologietechnologie, Fachhochschule Campus Wien, Sept. 2019.

[5] P Gröger. "Design, evaluation and verification of a small animal SPECT-CT imaging workflow for biologic adaptive radiotherapy". Master's thesis. MedTech, Fachhochschule Wiener Neustadt, Jan. 2021.

[6] L Hartl. "Energy ranges and spot map measurements in the frame of carbon ion beam commissioning at MedAustron". Bachelor's thesis. Atominstitut, Technische Universität Wien, Jan. 2020.

[7] S Haupt. "Dose Area Product Measurements with a Novel Large Area Ionisation Chamber in Scanned Proton Beams". Master's thesis. Atominstitut, Technische Universität Wien, Mar. 2019.

[8] N Leidenfrost. "Performance Analyse und technische Kommissionierung eines Kleintier-SPECT-Systems für den Einsatz in der biologisch adaptiven Strahlentherapie". Bachelor's thesis. Radiologietechnologie, Fachhochschule Wiener Neustadt, June 2021.

[9] B Nowak. "Performance Analyse und technische Kommissionierung eines Kleintier-CT-Systems für den Einsatz in der biologisch adaptiven Strahlentherapie". Bachelor's thesis. Radiologietechnologie, Fachhochschule Wiener Neustadt, June 2021.

[10] P Platzer. "Range Measurements for Carbon Ions in Water – Support during Commissioning Shifts at MedAustron". Bachelor's thesis. Radiologietechnologie, Fachhochschule Campus Wien, Sept. 2019.

[11] I Preineder. "Performance Analyse und technische Kommissionierung eines Kleintier-PET-Systems für den Einsatz in der biologisch adaptiven Strahlentherapie". Bachelor's thesis. Radiologietechnologie, Fachhochschule Wiener Neustadt, June 2021.

[12] NM Sprengnagel. "Dosimetrische Validierungsmessung für den Kohlenstoffstrahl im nicht-klinischen Forschungsraum bei MedAustron". Bachelor's thesis. Radiologietechnologie, Fachhochschule Campus Wien, May 2020.

[13] C Wculek. "Spotcharakteristika von Kohlenstoffionen in der Partikeltherapie". Bachelor's thesis. Radiologietechnologie, Fachhochschule Campus Wien, May 2020.

1.1 Intrafraction Adaptive Radiation Therapy

Intrafraction Surveillance Optimisation

Since 2019, several steps were tackled to optimise intrafraction surveillance and to reduce intrafraction motion in patients. A detailed analysis of patient imaging data with respect to intrafraction motion was presented by Zechner et al. [4]. Surface imaging data acquisition could be started only in 2021 after completing the clinical commissioning by the end of 2020. The analysis of clinical data is ongoing and will be preliminary completed by the end of 2021, with a focus on patients requiring either special attention during positioning or having tumours located in a moving body region. A potential improvement of the robot accuracy was worked out in the frame of a Master's thesis [9] but could, so far, not been implemented due to other priorities from the vendor of the robot, i.e. Buck Engineering and Consulting (BEC).

Markerless Real-Time Tumour Motion Surveillance

Software development on the 2D/3D image registration algorithm including bone suppression was finalised and validated by phantom measurements. All planed projects to correlate information on internal- and external motion, to perform on-line tumour imaging, and to define accelerator interface specifications for enabling triggered beam delivery were removed or postponed to the next research period. The main reason for this decision was the upcoming clinical focus to treat moving targets with (ventilated) breath hold or free breathing, which in turn requires only a simple interface in contrast to gating. The extraction of simple motion signals in order to start and stop the beam, is current work in progress but will only be operational in the research period 2022 – 2024.

Four-Dimensional Dose Calculation and Beam Delivery Optimisation

The human resources for this work package could be consolidated by the PhD student Franciska Lebbink, who is supported by a research grant funded by the Lower Austrian Science Call Dissertation. The validation of the dose calculation for protons for simple tumour movements was published in 2020 [3]. Since October 2020, four-dimensional computed tomography (4DCT) based dose tracking became possible for clinical patients due to the availability of required software tools, i.e. 4D dose tracking in the RayStation TPS, 4DCT imaging data, and motion signals from the surface scanner. In this context, a study on the retrospective analysis of clinical pancreas irradiation based on log-files was conducted. The analysis of the time resolved accumulated dose delivery based on accelerator log-files was addressed in a dedicated Bachelor's project by Thomas Schabhüttl [8]. The optimisation of beam delivery parameters for 4D ion-beam therapy started with a delay due to missing clinical data and was performed for selected indications in the frame of an ongoing Master's project. Within this research period, it was not possible to dosimetrically validate optimal beam delivery parameters. However, measurements for identifying tumour, lung, and rib motion amplitudes in need of motion mitigation were performed for protons and carbon ions. Thus, the interplay effect and range variations could be tackled using the advanced radiation dosimetry system (ARDOS) and the 4D dose tracking software. The degradation of the Bragg peak caused by homogeneous and heterogeneous lung tissue substitutes was investigated in the frame of a Master's thesis [5] and published in 2020 [1]. All initially planned projects on the improvement and validation of 4D imaging protocols needed to be removed from the research schedule due to a lack of human resources and the re-prioritisation of ongoing projects.

End-To-End Testing

Although the content related work was scheduled to start in 2021, it was decided to adapt and delay the start of this work package for the following reasons. Firstly, there was a strategic decision by EBG MedAustron to postpone the implementation of motion mitigation techniques and secondly, the additional PhD student Jacob Brunner, who is funded via the European Commission (EC) Intensive Training Network (ITN) project "Real Time Adaptive Particle Therapy of Cancer (RAPTOR)" only started in October 2021. His PhD project will concentrate on end-to-end testing for adaptive radiation therapy (ART). Consequently, research and development for end-to-end testing can then be extended to include intra- and interfractional moving targets in the upcoming research period with the support of two PhD students, i.e. Franciska Lebbink and Jacob Brunner, the new PhD student funded via the RAPTOR ITN project.

Eye Treatment Solution for MedAustron

The Austrian Research Promotion Agency (FFG) funded "Image Guided Particle Therapy of Uveal Melanoma with a Multiport Beam Arrangement (IMPACTOR)" project started officially in October 2019, but the core research personnel funded via this project started their work in the first and second half of 2020, respectively. Generally, all planned tasks could be finished according to the timeline of the project although the project started later than originally anticipated. The IMPACTOR project and its FFG funding will be continued until March 2022. Further funding applications are planned in the context of clinical implementation and validation.

At the beginning, a comprehensive literature review on eye treatment solutions at other ion-beam centres was performed. Next, Monte Carlo (MC) simulations were carried out to optimise the beam line for the envisaged eye treatment solution at MedAustron based on parameters motivated by the literature review. From a dosimetric point of view, similar parameters as dedicated eye beam lines seem to be feasible and respective experiments confirmed the MC simulations. Several detectors were experimentally tested for small field absolute dose measurements or relative dosimetry with respect to volume averaging and beam quality effects. Radiochromic films appeared to be a time efficient and valid two-dimensional dose detector. Therefore, a calibration procedure considerably faster than the current standard operating procedure (SOP) was established enabling frequent film calibrations. With regard to treat small tumours in the eye at shallow depths, a final treatment snout was designed based on the experience gained with the prototype and by additional MC simulations. The first hardware for the in-house developed eye treatment solution for validation under clinical conditions was produced in July 2021 and the final beam parameters are to be determined in the last quarter of 2021. A TPS beam model including the ocular snout was established in RayStation 10A based on the IR 1 commissioning data. The beam model was benchmarked against MC simulations in June 2021 and will be verified against experimental data in the end of 2021.

After defining the design specifications, a first prototype of the eye tracking hardware was finalised early 2021 by MedAustron. An improved version was then developed after mechanical tests and finalised mid 2021. At the same time, the eye tracking software was finalised. In a subsequent step, the interface to the MedAustron gating system was established and first gating tests were carried out in IR 1 in autumn 2021. A translation of the research outcome carried out in IR 1 to the clinical implementation in irradiation room 2 (IR 2) will be performed in 2022.

Scientific Output

4 Publications

[1] A Hranek, AF Resch, D Georg, and B Knäusl. “Investigation of the Bragg peak degradation caused by homogeneous and heterogeneous lung tissue substitutes: proton beam experiments and comparison to current clinical dose calculation”. In: *Physics in Medicine and Biology* 65.24 (Dec. 2020), page 245036.

[2] N Kostiukhina, H Palmans, M Stock, D Georg, and B Knäusl. “Dynamic lung phantom commissioning for 4D dose assessment in proton therapy”. In: *Physics in Medicine and Biology* 64.23 (Nov. 2019), page 235001.

[3] N Kostiukhina, H Palmans, M Stock, et al. “Time-resolved dosimetry for validation of 4D dose calculation in PBS proton therapy”. In: *Physics in Medicine and Biology* 65.12 (June 2020), page 125015.

[4] A Zechner, I Ziegler, E Hug, C Lütgendorf-Caucig, and M Stock. “Evaluation of the inter- and intrafraction displacement for head patients treated at the particle therapy centre MedAustron based on the comparison of different commercial immobilisation devices”. In: *Zeitschrift für Medizinische Physik* (2021), in Press.

5 Finalised Theses: 1 BSc, 3 MSc, 1 PhD

[5] A Hranek. "Experimental Validation of the Ripple Filter Effect of Lung Tissue Equivalent Materials in a Proton Beam". Master's thesis. Atominstitut, Technische Universität Wien, Oct. 2019.

[6] N Kostiukhina. "4D framework implementation towards investigating ion beam treatment techniques for intra-fractionally moving targets". PhD thesis. Department of Radiation Oncology, Medical University of Vienna, Mar. 2020.

[7] M Manni. "Simulation of different spot sizes for proton and carbon beams for medical purposes using GATE". Master's thesis. ISIMA, Clermont Auvergne, France, Oct. 2020.

[8] T Schabhüttl. "Accelerator log-file based dose rate extraction and the influence of dose rate variations on the 4D dose output". Bachelor's thesis. Technische Universität Wien, June 2021.

[9] C Stocker. "Optimierung der geometrischen Genauigkeit des Roboter-Patientenpositionierungssystems mit integriertem Tracking-System". Master's thesis. Technische Universität Kaiserslautern, Feb. 2019.

1.2 Interfraction Adaptive Radiation Therapy

Quantitative Evaluation of mpMRI for Clinical Pilot Studies

The focus within this work package was on deep learning and diffusion-weighted (DWI) magnetic resonance imaging (MRI) for head and neck patients. The upgrade of the MRI scanner could be successfully completed, however no patients received a dynamic contrast enhanced (DCE) scan for now. A dedicated multi-parametric magnetic resonance imaging (mpMRI) QA could not be implemented due to limited human resources and changed clinical prioritisation. It has to be highlighted that while a dedicated mpMRI phantom is of high interest, it is not a prerequisite to acquire clinical mpMRI images. In effect, hardly any clinic will have such a dedicated mpMRI phantom in use, which limits comparisons between different scanners on a quantitative basis.

About 30 head and neck patients received DWI MRI at four different time points, these data were evaluated in the MSc thesis of Alexander Kampitsch [12]. A dedicated ethic application for a clinical study seems not required, as the extended imaging protocol, i.e. mpMRI, is covered by the "register study", which is signed by nearly all patients. Regarding deep learning, good progress was achieved which is underlined by two publications [2, 3] and a finalised PhD thesis of Lukas Fetty, which is cited in Section 1.4 "Magnetic Resonance Guided Particle Therapy".

Evaluation of Clinical mpMRI Study and CBCT Supported ART Workflow

To fully tackle this work package, the funding of an EC project was defined as prerequisite, which was not granted. Thus, no human resources were available to conduct this research topic.

PSQA Based on Monte Carlo Dose Calculation

Monte Carlo based independent dose calculation is in clinical use for the horizontal and vertical proton beam line. With April 2021, this already reduced the patient-specific quality assurance (PSQA) time by 50 %. Clinical commissioning for the carbon-ion beam line is still

ongoing. The concept was also published [4, 8] and investigated in the Master's thesis of Lukas Scheuchenpflug [15]. A further manuscript was published presenting the Geant4 application for tomographic emission (GATE)-RTion-based independent dose calculation (IDEAL) for ion-beam therapy [5].

Furthermore, non-clinical research beam time was used to improve the beam model with special focus on lateral spread beam fragments. Large area ionisation chambers are in the centre of this research topic, resulting in two publications [6, 7].

Particle Therapy PET

Despite the valid ethics committee approval for a PET study with 15 patients suffering from meningioma, glioma, or chordoma, no patients were recruited in 2019. The plan to use the RayStation TPS to calculate the β^+-emitter distribution was not further pursued by MedAustron. In 2020, the PET scanner of MedAustron was sold following an adaption of clinical priorities and consequently the work package had to be stopped within the non-clinical research plan.

Scientific Output

9 Publications

[1] M Bolsa-Ferruz, H Palmans, D Boersma, M Stock, and L Grevillot. "Monte Carlo computation of 3D distributions of stopping power ratios in light ion beam therapy using GATE-RTion". In: *Medical Physics* 48.5 (2021), pages 2580–2591.

[2] L Fetty, M Bylund, P Kuess, et al. "Latent Space Manipulation for High-Resolution Medical Image Synthesis via the StyleGAN". In: *Zeitschrift für Medizinische Physik* 30.4 (Nov. 2020), pages 305–314.

[3] L Fetty, T Löfstedt, G Heilemann, et al. "Investigating conditional GAN performance with different generator architectures, an ensemble model, and different MR scanners for MR-sCT conversion". In: *Physics in Medicine and Biology Biology* 65.10 (May 2020), page 105004.

[4] L Grevillot, DJ Boersma, H Fuchs, et al. "Technical Note: GATE-RTion: a GATE/Geant4 release for clinical applications in scanned ion beam therapy." In: *Medical Physics* 47 (Aug. 2020), pages 3675–3681.

[5] L Grevillot, DJ Boersma, H Fuchs, et al. "The GATE-RTion/IDEAL Independent Dose Calculation System for Light Ion Beam Therapy". In: *Frontiers in Physics* 9 (Aug. 2021), page 424.

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[9] L Zimmermann, E Faustmann, C Ramsl, D Georg, and G Heilemann. "Technical Note: Dose prediction for radiation therapy using feature-based losses and One Cycle Learning". In: *Medical Physics* 48.9 (2021), pages 5562–5566.

6 Finalised Theses: 0 BSc, 4 MSc, 2 PhD

[10] PA Andrzejewski. "Multimodal imaging for radiation therapy planning and treatment response assessment". PhD thesis. Department of Radiation Oncology, Medical University of Vienna, Oct. 2019.

[11] M Daniel. "Making multiparametric imaging radiation-oncology ready". PhD thesis. Department of Radiation Oncology, Medical University of Vienna, Sept. 2019.

[12] A Kampitsch. "Quantitative Analyse der Apparent Diffusion Coefficient-Karten von Teilchentherapie-HirntumorpatientInnen in Abhängigkeit der applizierten Dosis". Master's thesis. Studiengang Medizintechnik, Campus Linz, Fachhochschule Oberösterreich, June 2021.

[13] M Matthewman. "Investigating the potential of generative adversarial networks (GANs) for autosegmentation in radiation oncology". Master's thesis. Atominstitut, Technische Universität Wien, Sept. 2021.

[14] VA Moser. "Characterization of the IBA StingRay plane-parallel ionization chamber for proton and carbon dosimetry". Master's thesis. Atominstitut, Technische Universität Wien, Oct. 2021.

[15] L Scheuchenpflug. "Development of a commissioning tool for the independent dose calculation software IDEAL for the patient specific quality assurance at MedAustron". Master's thesis. Universität Wien, Aug. 2020.

1.3 Imaging with Ion Beams

The main objective of this project was to develop a functional proton computed tomography (pCT) demonstrator, test it in beam and perform an image reconstruction of a three-dimensional stopping power map of a small object. The project was subdivided into three strongly integrated work packages and, in addition, a contribution to the development of a new beam delivery system was planned.

Monte Carlo Simulation and Dose Estimation

For planning and optimisation purposes, the experimental work was accompanied by Monte Carlo simulations of the tracking and the energy measurement part of the imaging system. In this respect, major resources were dedicated to the modelling of the range telescope used for residual energy measurement. It was modelled in Geant4 in the scope of a Master's thesis [15] and had to be substantially upgraded in terms of hardware and operating software, supported by a Bachelor's thesis [13] and Master's thesis [9]. Owing to that, less activities with respect to dosimetry were performed and were postponed to upcoming research periods.

Proton Computed Tomography Set-Up and Data Taking

Following the modelling and optimisation process using Geant4 simulations, the final version of the demonstrator was built and tested in five consecutive beam time shifts in October 2020. It constituted of double-sided silicon strip detectors available at the Institute of High Energy Physics (HEPHY) for the tracking part and the range telescope for residual energy measurement of ions traversing the object. The technological progress was reported at several conferences and workshops and was published in 2020 [1, 5].

The successful testing of the demonstrator end of 2020 allowed to focus on testing new detector technology for particle tracking and residual energy measurement in a future pCT demonstrator in the remaining beam times in the research period 2019 – 2021. Concluding

from these tests and supported by a statement given by experts in the field, i.e. G. Dedes (LMU, Germany) and R. Schulte (LLU, USA), low gain avalanche detectors (LGADs) providing excellent timing properties were identified to be used as tracking detectors as well as in a calorimeter based on time-of-flight (TOF) energy measurement. The required hardware for the LGAD based demonstrator is planned to be bought from funds available at the Technische Universität (TU) Wien. The project will be implemented in the research period starting from 2022 as a continuing effort to lay the foundation for ion-beam imaging in the clinics using new sensor technology.

As prerequisite for pCT with available detectors and electronics, substantial effort was undertaken to reduce the nominal particle flux at the research facility MedAustron. A dedicated set-up and software were developed to enable single particle counting in IR 1, since the beam diagnostics in the particle accelerator beam line cannot determine fluxes below a certain threshold. Parts of this work were performed in the scope of a Bachelor's thesis [14] and were eventually fully reported in a publication [6]. The released low flux settings for proton beams are now available to all users.

The two PhD theses performed in the scope of the ion-imaging project (A. Burker and F. Ulrich-Pur) are planned to be finished by the beginning of 2022.

Stopping Power Reconstruction

For image reconstruction, a collaboration with imaging experts from the Centre for Medical Physics and Biomedical Engineering, Medical University of Vienna could be established and the open-source cone-beam CT software framework tomographic iterative GPU-based reconstruction toolbox (TIGRE) was identified to be applicable to the pCT reconstruction problem. Its first application to the pCT reconstruction problem was established in a Master's thesis [10], further investigated in a Bachelor's thesis [12] and later published [2]. Progress in this work package was enabled by a PhD student funded by the FFG in 2019 (S. Kaser). Recently, another manuscript for optimising the estimation of the most likely path of an ion in the traversed object during image reconstruction was published [4].

Beam Delivery System Development

Work on the beam delivery system was stopped in the scope of the ion beam imaging project in 2019 and went ahead separately, driven by MedAustron.

Scientific Output

6 Publications

[1] A Burker, T Bergauer, A Hirtl, et al. "Imaging with Ion Beams at MedAustron". In: *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* 958 (Apr. 2020). 15th Vienna Conference on Instrumentation (VCI), Vienna, Austria, Feb 18-22, 2019.

[2] S Kaser, T Bergauer, W Birkfellner, et al. "First application of the GPU-based software framework TIGRE for proton CT image reconstruction". In: *Physica Medica* 84 (Apr. 2021), pages 56–64.

[3] S Kaser, T Bergauer, A Burker, et al. "Different radiographic imaging modalities with a proton computed tomography demonstrator". In: *Journal of Instrumentation* (2021). 22nd International Workshop on Radiation Imaging Detectors, Accepted Author Manuscript.

[4] S Kaser, T Bergauer, A Hirtl, et al. "Calculating $1/\beta^2 p^2$ for most likely path estimates for protons and helium ions using an analytical model". In: *Physica Medica* 89 (Sept. 2021), pages 169–175.

[5] F Ulrich-Pur, T Bergauer, A Burker, et al. "Imaging with protons at MedAustron". In: *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* 978 (Oct. 2020), page 164407.

[6] F Ulrich-Pur, L Adler, T Bergauer, et al. "Commissioning of low particle flux for proton beams at MedAustron". In: *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* 1010 (2021), page 165570.

10 Finalised Theses: 5 BSc, 5 MSc, 0 PhD

[7] M Christanell. "4H-Silicon carbide as real time monitor for high-intensity ion beams". Master's thesis. Institut für Hochenergiephysik, Technische Universität Wien, Sept. 2021.

[8] V Dangl. "Evaluation of an FPGA-Module for the application in a particle detector readout system". Bachelor's thesis. Institut für Hochenergiephysik, Technische Universität Wien, July 2021.

[9] B Huber. "Development and assessment of a calorimeter data acquisition and evaluation software for ionizing particles". Master's thesis. Wien, 2020.

[10] S Kaser. "Geant4 Simulations for Ion CT Image Reconstruction". Master's thesis. Atominstitut, Technische Universität Wien, Oct. 2019.

[11] J Lang. "GATE simulation of an adaptable proton CT setup". Bachelor's thesis. Atominstitut, Technische Universität Wien, Oct. 2021.

[12] M Langschwert. "Quality Factors for pCT Image Reconstruction". Bachelor's thesis. Technische Universität Wien, 2021.

[13] P Maier. "Calibration of a calorimeter for proton computed tomography". Bachelor's thesis. Technische Universität Wien, June 2020.

[14] P Sommerer. "Implementation of a low particle flux rate monitor for the development of a protonCT set-up". Bachelor's thesis. Technische Universität Wien, Feb. 2020.

[15] V Teufelhart. "Geant4 simulation of a particle range telescope used as calorimeter at MedAustron". Master's thesis. Atominstitut, Technische Universität Wien, July 2020.

[16] M Tomaschek. "Material study on the use of silicon carbide in position-sensitive particle detectors". Master's thesis. Atominstitut, Technische Universität Wien, Sept. 2021.

1.4 Magnetic Resonance Guided Particle Therapy

This research project successfully allowed to keep our group at the frontier of the rapidly evolving field of magnetic resonance (MR) guided particle therapy [3].

An integral part of this research period was the Austrian Science Fund (FWF) project MAGIC-PRO, successfully completed end of 2020, resulting in the finalisation of two PhD theses [12, 14]. This research topic will be continued in the next research period. To complement our manpower for the next research period, additional efforts to secure third party funding are currently ongoing.

Magnetic Resonance Workflow Development

A promising magnetic resonance workflow was developed focusing on a conversion method from MR images to synthetic CTs. This workflow was based on a deep learning network using conditional generative adversarial networks (GANs). Evaluations performed using patient data, acquired using a CT and a low-field open MR scanner installed at the Medical University of Vienna and applied to photon and proton treatment plans indicate clinical acceptable results [10]. Furthermore, an MRI sequence independent synthetic computed tomography (sCT)-generator was implemented and validated with proton therapy patients [11].

Dose Calculation Development and Implementation into TPS

To allow treatment planning for MR guided particle therapy, a multi-stage approach was adopted. Firstly, a MC method of our beam line including our experimental research magnet was commissioned to be used for benchmarking [5]. Secondly, after developing an accurate dose calculation algorithm suitable for magnetic fields, it was implemented into the research treatment planning system MatRad and successfully validated. This allowed us to perform a proof-of concept treatment planning study indicating the feasibility to perform treatment planning in magnetic field environments [7]. Using in vitro biological experiments irradiating 2D cell cultures with protons and carbon ions in magnetic fields, we could rule out additional biological effects due to the presence of magnetic fields [4].

Development of Dosimetry Framework and Detector Characterisation

Being able to perform accurate dosimetry is a key aspect of future MR guided particle therapy. Using our 1 T experimental magnet, dosimetric measurements in proton beams were performed helping to characterise the response of clinically employed detectors. In a first step, passive Gafchromic EBT3 films were investigated, showing no magnetic field dependency [6, 9]. In a second step, a set of five detectors [2] was investigated, exhibiting a small but significant magnetic field dependent change in response behaviour. This was assisted by the development of a computer-controlled water phantom suitable to work in the constrained environment of our research magnet performed by a MSc student [1, 13]. Ongoing studies investigate the detector response during carbon-ion irradiation. Furthermore, a cooperation with the University of Oldenburg (Germany) is planned for 2022 to further investigate magnetic field influences. In addition, a cooperation with the Centre for Radiation Research in Oncology (OncoRay) in Germany was initiated in 2021. The cooperation aims to establish a dosimetry workflow and QA process for MR guided proton therapy.

Scientific Output

10 Publications

[1] H Fuchs, F Padilla-Cabal, A Hummel, and D Georg. “Technical Note: Design and commissioning of a water phantom for proton dosimetry in magnetic fields”. In: *Medical Physics* 48.1 (Jan. 2021), pages 505–512.

[2] H Fuchs, F Padilla-Cabal, L Zimmermann, H Palmans, and D Georg. “MR-guided proton therapy: Impact of magnetic fields on the detector response”. In: *Medical Physics* 48.5 (May 2021), pages 2572–2579.

[3] A Hoffmann, B Oborn, M Moteabbed, et al. "MR-guided proton therapy: a review and a preview". In: *Radiation Oncology* 15.1 (2020), page 129.

[4] S Kerschbaum-Gruber, F Padilla-Cabal, E Mara, et al. "An external magnetic field does not influence survival and DNA damage after proton and carbon ion irradiation in human cancer cells". In: *Zeitschrift für Medizinische Physik* (2021), Accepted Author Manuscript.

[5] F Padilla-Cabal, J Alejandro Fragoso, AF Resch, D Georg, and H Fuchs. "Benchmarking a GATE/Geant4 Monte Carlo model for proton beams in magnetic fields". In: *Medical Physics* 47.1 (Jan. 2020), pages 223–233.

[6] F Padilla-Cabal, P Kuess, D Georg, et al. "Characterization of EBT3 radiochromic films for dosimetry of proton beams in the presence of magnetic fields". In: *Medical Physics* 46.7 (July 2019), pages 3278–3284.

[7] F Padilla-Cabal, AF Resch, D Georg, and H Fuchs. "Implementation of a dose calculation algorithm based on Monte Carlo simulations for treatment planning towards MRI guided ion beam therapy". In: *Physica Medica: European Journal of Medical Physics* 74 (June 2020), pages 155–165.

[8] AF Resch, H Palmans, D Georg, and H Fuchs. "The practical radius of a pencil beam in proton therapy". In: *Zeitschrift fü Medizinische Physik* 31.2 (2021). Special Issue: Ion Beam Therapy, Part I, pages 166–174.

[9] AF Resch, PD Heyes, H Fuchs, et al. "Dose- rather than fluence-averaged LET should be used as a single-parameter descriptor of proton beam quality for radiochromic film dosimetry". In: *Medical Physics* 47.5 (May 2020), pages 2289–2299.

[10] L Zimmermann, M Buschmann, H Herrmann, et al. "An MR-only acquisition and artificial intelligence based image-processing protocol for photon and proton therapy using a low field MR". In: *Zeitschrift für Medizinische Physik* 31.1 (Feb. 2021), pages 78–88.

[11] L Zimmermann, B Knäusl, M Stock, et al. "Using an MRI sequence independent Convolutional Neural Network for synthetic head CT generation in proton Therapy". In: *Zeitschrift für Medizinische Physik* (2021), Accepted Author Manuscript.

3 Finalised Theses: 0 BSc, 1 MSc, 2 PhD

[12] L Fetty. "Magnetic Resonance Imaging only treatment planning for photon and particle therapy". PhD thesis. Department of Radiation Oncology, Medical University of Vienna, June 2020.

[13] A Hummel. "Development of a water phantom for a high magnetic field environment in a constrained volume". Master's thesis. Studiengang Medizintechnik, Campus Linz, Fachhochschule Oberösterreich, June 2019.

[14] F Padilla-Cabal. "A dose calculation platform for treatment planning towards magnetic resonance image guided particle therapy". PhD thesis. Department of Radiation Oncology, Medical University of Vienna, Mar. 2020.

1.5 Energy Transfer Mechanisms and Applications in Biology and Physics

Method Establishment

Based on the experience from the previous research period 2016 – 2018, the radiobiology set-up was adapted. Instead of the solid phantom [2], a new water-phantom-based experimental set-up was implemented [1]. The programme of experimental microdosimetry in the period from 2019 until 2021 investigated all the types of microdosimeters available today: the synthetic

diamond microdosimeters, the silicon diodes, and the tissue equivalent proportional counters (TEPCs). While characteristics and use of diamond microdosimeters were well known by the microdosimetry group before the programme began, silicon microdosimeters and TEPCs were newly purchased. Upgraded versions of the solid-state detectors allowed moving the measurements from a water-equivalent plastic phantom to the more realistic and versatile water phantoms. Custom-made set-ups were studied and implemented to allow synchronised collection of microdosimetric spectra with different detectors. Proton and carbon-ion beams at different energies were investigated as single beams, scanned beams, or beams modulated to form spread-out Bragg peaks (SOBPs). For almost all the specific conditions and positions investigated, two detector types were used.

Model Development for Microdosimetry

The goal of the microdosimetry group is dual. From one side, completing the characterisation of the microdosimeters in ion beams; this will be done mirroring, in a smaller scale, the procedures used for implementing dosimetry in ion-beam therapy. From the other side, the effort will be to evolve from the restricted area of microdosimetry research to the wider field of microdosimetric applications in radiobiology and clinics. The so far established methodology allows collecting reproducible experimental outcomes and providing univocal theoretical interpretations of the results of various available microdosimeters.

These experimental results constitute a considerable set of data for the comparison of the response from different detectors. This is the key element for redefining methodology and microdosimetry algorithms to link physics data with radiobiological data for ion-beam research. In the context of commissioning the new water phantom, the effective point of measurement (P_{eff}) of the ionisation chamber was investigated and refined for proton and carbon-ion beams. P_{eff} is especially important for radiobiological or detector response experiments at the distal fall-off, due to the high linear energy transfer (LET). In microdosimetry, new protocols for the linearisation of the response of the electronic chain, the calibration of the microdosimetric spectra, the evaluation of the contribution of energy-loss straggling and delta-ray escape, as well as the impact of the geometrical characteristics of the detector on the spectra were established. This setting represents the experimental basis, which is used to obtain detector-independent information on the radiation quality. So far, the most commonly applied physics parameter, i.e. the dose-averaged LET, is a result of pure computations and there is no direct comparison with measurements, neither in radiobiological studies, nor during commissioning of the irradiation facilities, nor in regular quality assurance procedures. Based on the renew formalism, the probability distributions of the LET was obtained, indirectly, from the microdosimetric measurements. The mean values of LET in water targets can be readily used in clinics.

An ongoing goal concerns the microdosimetric mapping of the irradiation field used in cell irradiation, allowing pairing the biological and physical outcomes in specific points. This innovative procedure is also intended to support radiobiological comparisons in different ion-beam facilities, an activity that is foreseen in the framework of the project "Heavy Ion Therapy Research Integration *plus* (HITRI*plus*)", which is funded by the EC and started in April 2021.

A clear evidence emerged from the experimental and the methodological studies about the inadequacy of current definition of lineal energy when ion beams are investigated. This topic was brought to the international arena, urging the need of reconsidering and adapting the definition of lineal energy.

Ion-Beam Irradiation Effects in Immunologically Hot Tumours

Research with hot tumour indications largely focused on head and neck squamous cell carcinoma (HNSCC) cells and non-small cell lung carcinoma (NSCLC) cells. Both tumour types display relatively high numbers of mutations and hence are classified as "hot" or "inflamed" phenotypes. Due to the high mutational load, hot tumours may be sensitive to targeted treatments including deoxyribonucleic acid (DNA) repair inhibition, specifically when combined with particle treatment. This work package assessed the ability of Ganetespib and Triapine treatments to radiosensitise HNSCC and NSCLC. Ganetespib, a heat-shock-protein 90 inhibitor (HSP90i), was found to be superior to Triapine, a ribonucleotide reductase inhibitor. Subsequent proton experiments hence focused on Ganetespib only. Used in a clinically relevant 2 nM concentration, Ganetespib effectively suppressed DNA damage repair-relevant RAD51 and thus interfered with repair of radiation-induced DNA damage. RAD51 expression was enhanced after proton irradiation as compared to photons for the same physical dose. Thus, HSP90i was identified as a suitable strategy to sensitise HNSCC and NSCLC to specifically proton irradiation, exploiting proton-specific RAD51 dynamics.

Ion-Beam Irradiation Effects in Immunologically Cold Tumours

The second main radiobiology focus is radiation-induced anti-tumour immunogenicity, i.e. developing strategies to convert immunologically cold (unresponsive) tumour types into immunologically hot (responsive) tumours. This work package focuses on triple-negative breast cancer (TNBC) and pancreatic cancer (PC). Radiation appears to be an ideal neoadjuvant to immunotherapy to improve immunotherapy efficacy in hot tumour types but data for cold tumours is scarce. In order to establish a collaboration with experts of the field, Sylvia Kerschbaum-Gruber joined the lab of Prof. Sandra Demaria for six months in 2019 [13]. Collaborative efforts are ongoing ever since with a focus on TNBC. Three-dimensional tumour spheroid models were established and hypoxia was identified as a potent inhibitor of radiation-induced anti-tumour immunogenicity. Different radiation regimen and combinational approaches were tested with a focus on PC. Fractionated irradiation (3×8 Gy) was identified to elicit an immunogenic signalling response, whereas single dose irradiation did not (Master's theses of K. Stasny [24] and L. Komornik [19], Bachelor's thesis of C. Faustbeck [17]). Our results revealed that protons are not superior to reference X-rays for any endpoint investigated. Pancreatic cancer cells, as compared to TNBC cells, were found to be even less responsive. As potential culprit dysfunctional STING activation (Master's thesis of A. Falkner [16]) following radiation-induced accumulation of cytosolic dsDNA (Bachelor's theses of N. Kurikkattil [20] and S. Vadakumcheril [25]) was identified. Combinational treatment strategies with DNA repair inhibitors Olaparip (poly ADP-ribose polymerase inhibitor (PARPi)) and KU-5593 (ataxia telangiectasia mutated inhibitor (ATMi)) were recently initiated to investigate a potential link between DNA repair and radiation-induced anti-tumour immunogenicity. ATMi in combination with a radiotherapy dose that does not elicit an immunogenic response on its own, did achieve significant immunogenic signalling. PARPi, in comparison, is less potent. ATMi plus radiotherapy may be a strategy to convert pancreatic cancer into an immunogenic cancer (Bachelor's thesis M. Rehberger [23]). In addition to investigating potential stimulation of immunogenic signalling in TNBC and PC, chondrosarcoma research was becoming integral part of this research avenue, enabled by an FWF grant (Birgit Lohberger and Dietmar Georg, FWF). No immunogenic signalling could be elicited

with neither X-rays, nor protons or carbon ions. Emphasis was hence directed towards other therapy-relevant signalling and DNA repair modulation combined with irradiation was identified as potential treatment strategy [5]. So far, protons were not found to be superior to X-rays for any endpoint investigated. Combinational treatment with protons and ATMis/ataxia telangiectasia and Rad3-related protein inhibitors (ATRis) was found to induce synthetic lethality. Carbon ions in combination with ATMis/ATRis achieved best results.

Scientific Output

13 Publications

[1] S Barna, A Resch, M Puchalska, D Georg, and H Palmans. "Technical Note: Experimental determination of the effective point of measurement of the PTW-31010 ionization chamber in proton and carbon ion beams". In: *Zeitschrift für Medizinische Physik* (2021), Accepted Author Manuscript.

[2] M Clausen, S Khachonkham, S Gruber, et al. "Phantom design and dosimetric characterization for multiple simultaneous cell irradiations with active pencil beam scanning". In: *Radiation and Environmental Biophysics* 58 (Sept. 2019).

[3] P Colautti, G Magrin, H Palmans, MA Cortés-Giraldo, and V Conte. "Characterizing radiation effectiveness in ion-beam therapy Part II: Microdosimetric detectors". In: *Frontiers in Physics* 8 (Oct. 2020), page 451.

[4] S Khachonkham, E Mara, S Gruber, et al. "RBE variation in prostate carcinoma cells in active scanning proton beams: In-vitro measurements in comparison with phenomenological models". In: *Physica Medica* 77 (2020), pages 187–193.

[5] B Lohberger, D Glaenzer, N Eck, et al. "Effects of a combined therapy of bortezomib and ionizing radiation on chondrosarcoma three-dimensional spheroid cultures". In: *Oncology Letters* 21.6 (Mar. 2021), page 428.

[6] G Magrin, C Verona, M Ciocca, et al. "Microdosimetric characterization of clinical carbon-ion beams using synthetic diamond detectors and spectral conversion methods". In: *Medical Physics* 47.2 (Feb. 2020), pages 713–721.

[7] G Magrin, C Verona, G Verona-Rinati, and M Stock. "Microdosimetry of clinical ion beams: converting spectra from diamond slab to water of different shapes". In: *Radiation Protection Dosimetry* 183.1–2 (May 2019), pages 167–171.

[8] E Mara, M Clausen, S Khachonkham, et al. "Investigating the impact of alpha/beta and LET_d on relative biological effectiveness in scanned proton beams: an in vitro study based on human cell lines". In: *Medical Physics* 47.8 (Aug. 2020), pages 3691–3702.

[9] A Parisi, T Sato, Y Matsuya, et al. "Development of a new microdosimetric biological weighting function for the RBE10 assessment in case of the V79 cell line exposed to ions from ^{1}H to ^{238}U". In: *Physics in Medicine and Biology* 65.23 (Dec. 2020), page 235010.

[10] M Puchalska. "Modelling and measurements of distributions in an adult human phantom undergoing proton scanning beam radiotherapy: lung- and prostate-located tumours". In: *Radiation and Environmental Biophysics* 60.2 (Mar. 2021), pages 243–256.

[11] M Scholz, T Friedrich, G Magrin, et al. "Characterizing Radiation Effectiveness in Ion Beam Therapy Part I: Introduction and Biophysical Modeling of RBE Using the LEMIV". In: *Frontiers in Physics* 8 (Aug. 2020), page 272.

[12] C Verona, GAP Cirrone, G Magrin, et al. "Microdosimetric measurements of a monoenergetic and modulated Bragg Peaks of 62 MeV therapeutic proton beam with a synthetic single crystal diamond microdosimter". In: *Medical Physics* 47.11 (Nov. 2020), pages 5791–5801.

[13] E Wennerberg, S Spada, NP Rudqvist, et al. "CD73 Blockade Promotes Dendritic Cell Infiltration of Irradiated Tumors and Tumor Rejection". In: *Cancer Immunology Research* 8.4 (Apr. 2020), pages 465–478.

12 Finalised Theses: 6 BSc, 4 MSc, 2 PhD

[14] M Barenghi. "Microdosimetry modelling for proton therapy beams with GATE". Master's thesis. Faculty of Science and Technology, University of Milan, Sept. 2020.

[15] S Deycmar. "Differential Response to Photon versus Proton Radiotherapy – Biological Implications for New Indications and Combined Treatment Concepts". PhD thesis. Faculty of Science, University of Zurich, Mar. 2020.

[16] A Falkner. "Radiation-induced anti-tumor immunogenicity in pancreatic cancer in vitro models – photon- vs particle irradiation effects". Master's thesis. Master Molekulare Biologie, Universität Wien, June 2021.

[17] C Faustbeck. "Strahleninduzierte Veränderungen der Anti-Tumor-Immunantwort". Bachelor's thesis. Biomedizinische Analytik, Fachhochschule Wiener Neustadt, May 2021.

[18] S Khachonkham. "Improving Relative Biological Effectiveness (RBE) Modelling in Proton Beam Therapy: Medical Physics and Radiobiological Aspects". PhD thesis. Department of Radiation Oncology, Medical University of Vienna, Nov. 2019.

[19] L Komornik. "Strahleninduzierte Immunogenität in Pankreaskrebszellen – Einfluss von Photonen- und Partikelbestrahlung". Master's thesis. Massenspektrometrie und Molekulare Analytik, Fachhochschule Joanneum Graz, Aug. 2021.

[20] N Kurikkattil. "Radiation-Induced Immunogenic Key Signalling in MIA PaCa-2 Human Pancreatic Cancer Cells – Proton vs Photon Irradiation". Bachelor's thesis. Medical and Pharmaceutical Biotechnology, IMC Fachhochschule Krems, July 2021.

[21] M Lutz. "Regulierung von Anti-Tumor-Immunitätsmarker nach Photonen- und Partikelbestrahlung in einem multizellulären Tumor-Sphäroid-Modell". Bachelor's thesis. Biomedizinische Analytik, Fachhochschule Wiener Neustadt, May 2020.

[22] L Neuhauser. "Bestimmung des Interferon signalling in einem multizellulären Tumor-Sphäroid-Modell nach Photonen- und Partikelbestrahlung". Bachelor's thesis. Biomedizinische Analytik, Fachhochschule Wiener Neustadt, May 2020.

[23] M Rehberger. "Auswirkungen von DNA-Reparaturinhibitoren auf die strahleninduzierte Tumorimmunogenität in Pankreaskarzinomzelllinien". Bachelor's thesis. Biomedizinische Analytik, Fachhochschule Campus Wien, May 2021.

[24] K Stasny. "Radiation-Induced Anti-Tumor Immunogenicity in Pancreatic Cancer 2D Culture and a Multicellular Tumor Spheroid Model". Master's thesis. Medical and Pharmaceutical Biotechnology, IMC Fachhochschule Krems, Dec. 2020.

[25] S Vadakumcheril. "Radiation Induced Immunogenic Key Signaling in Panc-1 Human Pancreatic Cells: Photon vs Proton". Bachelor's thesis. Medical and Pharmaceutical Biotechnology, IMC Fachhochschule Krems, July 2021.

1.6 Pre-Clinical Animal Research

Commissioning of the Animal Facility at MedAustron

Regular meetings with veterinarian Dr. Filip took place since Q2 2021 to prepare all required infrastructure for animal housing. The final application was submitted to the ministry and is currently reviewed. A positive decision by the ministry is expected for November 2021.

In Vitro Tumour Transplantation Techniques

Until July 2019, researchers at the Medical University of Vienna (Jakob Kowaliuk, Verena Kopatz) worked on the generation of transplantable osteosarcoma (OS) tumour constructs as well as on similar constructs for pancreatic ductal adenocarcinoma (PDAC). By the end of 2019, an overall re-organisation of the research group "Applied and Translational Radiobiology (ATRAB)" at the Medical University of Vienna took place. The main topics of project "Pre-Clinical Animal Research" were changed in comparison to the "MedAustron Non-Clinical Research Strategy 2019 – 2020 – 2021" and the work package on "In Vitro Tumour Transplantation Techniques" was no longer pursued. As there were no shifts foreseen for this project, no further measures were taken.

Commissioning of Animal Irradiation Set-Ups

Dosimetric studies as part of a preparatory work for small-animal irradiation were performed with a dedicated phantom designed for horizontal particle beams. The irradiation workflow followed very closely typical patient workflow with CT scan of a phantom and CT import into RayStation TPS. Target and other structures were delineated and the dose was calculated with a MC calculation algorithm. Small-field irradiations were performed for different target sizes with the smallest investigated cylindrically-shaped target of 8 mm (diameter). Due to relatively high therapeutic beam energies, a pre-absorber (range shifter or bolus) was required to irradiate shallow targets typical for small animals. To shield the not targeted regions, collimators were included in the set-up. Different detectors, e.g. thermoluminescent detectors, Gafchromic EBT3 films and micro diamond detector, were employed to record the dose and dose distribution. Experimentally validated doses were in a good agreement with the prescribed doses including the collimator in the irradiation set-up with the range shifter for proton beams. One PhD and one MSc student were involved in this experiment.

In addition to the dosimetric studies, several activities around the design of the mouse bed, the positioning table as well as the collimator design in the YXLON were performed envisaging a transfer of the set-up for proton irradiation.

Effects of Ion Beams on Tumour Microenvironment Signalling and Tumour Reoxygenation

By the end of 2019, an overall re-organisation of the research group "Applied and Translational Radiobiology (ATRAB)" at the Medical University of Vienna took place. The main topics of project "Pre-Clinical Animal Research" were changed in comparison to the "MedAustron Non-Clinical Research Strategy 2019 – 2020 – 2021" and the work package on "Effects of Ion Beams on Tumour Microenvironment Signalling and Tumour Reoxygenation" was no longer pursued. The requested shifts for this project for 2020 and 2021 were not allocated, but partly invested into the main research and development activities in work package "Commissioning of Animal Irradiation Set-Ups", i.e. to establish the technical irradiation setting and an appropriate treatment planning workflow based in micro-CT imaging for small-animal irradiation.

PET/CT Imaging to Monitor Alterations in Tumour Biology

The main activities in this work package focused on the acceptance testing and commissioning of the small-animal imaging equipment, see also Section 1.0 "Treatment Planning and Imaging for Small-Animal Irradiation". Due to the re-organisation of the research group "Applied and

Translational Radiobiology (ATRAB)" at the Medical University of Vienna by the end of 2019, no further radiobiology driven activities took place. The requested shifts for this project for 2021 were not allocated, but partly invested into the main research and development activities in work package "Commissioning of Animal Irradiation Set-Ups", i.e. to establish the technical irradiation setting and an appropriate treatment planning workflow based in micro-CT imaging for small-animal irradiation.

Scientific Output

5 Finalised Theses: 4 BSc, 1 MSc, 0 PhD

[1] B Haderer. "Charakterisierung der Effektivität von Photonenstrahlen auf das Überleben von Pankreaskarzinomzellen". Bachelor's thesis. Biomedizinische Analytik, Fachhochschule Wiener Neustadt, May 2019.

[2] D Hatter. "Charakterisierung der Effektivität von Photonenstrahlen auf das Überleben von Osteosarkomzellen". Bachelor's thesis. Biomedizinische Analytik, Fachhochschule Wiener Neustadt, May 2019.

[3] B Kientzl. "Charakterisierung der Effektivität von Photonenstrahlen auf das Überleben von Osteoblasten". Bachelor's thesis. Biomedizinische Analytik, Fachhochschule Wiener Neustadt, Jan. 2019.

[4] K Secco. "Radiation-induced changes in the expression of p16/p21 in the oral mucosa of mice – the effect of heparin". Master's thesis. Massenspektrometrie und Molekulare Analytik, Fachhochschule Joanneum Graz, June 2019.

[5] H Tiefenbacher. "Dosimetric Preparations for Small Animal Irradiations with an X-Ray Beam". Bachelor's thesis. Medical and Pharmaceutical Biotechnology, IMC Fachhochschule Krems, May 2019.

1.7 Accelerator Physics

The project on "Accelerator Physics" was, not described in the last research programme, i.e. "MedAustron Non-Clinical Research Strategy 2019 – 2020 – 2021", since the leading professor of that project was appointed at the TU Wien only by the end of 2019. Thus the content related work started early in 2020 with three work packages requiring beam time.

Slow-Extraction Optimisation

Although the current used third order slow-resonant extraction technique at the MedAustron synchrotron is a widely used technique in many accelerator centres, it is not suitable for re-acceleration purposes. The time necessary to change the delivered beam energy is of the order of few seconds, because a whole new acceleration cycle has to be performed. Additionally, the intensity of the beam can only be modulated within tens of milliseconds due to the slow reaction of the Betatron core magnet. Thus, a feasibility study for implementing the so-called RF knock-out extraction method that potentially brings several advantages for operation was launched in 2020. In parallel to the theoretical analysis, the simulation results were continuously benchmarked with machine measurements. The investigations are mainly performed by a PhD student from the TU Wien, who started in October 2020, and are additionally embedded within a research agreement with Conseil Européen pour la Recherche Nucléaire (CERN), covering on a more general basis research on slow-resonant extraction techniques.

So far, tests were created to generate slow extraction methods using the constant optics slow extraction (COSE) and tune sweep methodologies performing the extractions, while keeping the beam bunched so that part of the beam could be extracted and then re-accelerated for a secondary extraction with a different energy. This topic will be continued in the research period 2022 – 2024.

Commissioning of Helium-Ion Beams

Following the common research interest for using helium ions as indicated already in the last two Research Strategies, the project for providing helium ions at MedAustron in IR 1 was launched in 2020. First preparation work was carried out in parallel to the regular beam operation in the injector hall, i.e. installation, configuration, and commissioning of the third source branch for helium ions. The electron cyclotron resonance ion source was integrated into the test system and commissioned remotely with integration into the MedAustron control system. The first start-up with plasma was achieved using all available gas types with particular emphasis on helium ions in a first phase. The further commissioning of the accelerator components especially in the synchrotron and the high-energy beam transport (HEBT) line into IR 1 is one of the main tasks for the upcoming years in the research area "Accelerator Physics".

Gantry Rotator Commissioning

The correct adaptation of the transversally asymmetric beam to the gantry, independent of the rotation angle, is a long-standing challenge of medical synchrotron centres. At MedAustron, an innovative solution using a rotator structure for the phase space rotation of the beam was implemented. In 2020, this worldwide unique rotator structure located in the HEBT line of the MedAustron accelerator complex was commissioned and preliminary acceptance and final acceptance measurements were performed. During the final acceptance tests, the behaviour of the rotator under non-ideal conditions, such as simulating drifts at the electrostatic septum of the extraction, angular and position errors at the entrance of the rotator, and interpolated angle settings for the rotator were investigated. All tasks were significantly supported by a Master's student from the TU Wien. Since the accelerator commissioning for the gantry operation was finalised in 2021, this topic has been closed and will not be continued in the upcoming research period

Scientific Output

3 Finalised Theses: 1 BSc, 1 MSc, 1 PhD

[1] F Kühteubl. "Design Study of Radio Frequency Knockout Slow Extraction for the MedAustron Synchrotron". Master's thesis. Atominstitut, Technische Universität Wien, Mar. 2020.

[2] P Langhans. "Time of flight analysis for protons and carbon ions in the Medium Energy Beam Transfer Line of MedAustron". Bachelor's thesis. Technische Universität Wien, Sept. 2020.

[3] A Wastl. "Tomographical reconstruction of the horizontal phase space of a 3rd order slow resonant extracted proton beam of the MedAustron synchrotron". PhD thesis. Atominstitut, Technische Universität Wien, Feb. 2020.

1.8 Regular Follow Up of Activities and Project Progress Supervision

As already introduced in the first research period, the regular Executive Committee meetings were continued in order to discuss with the project coordinators practical issues and the research progress in the different projects. Since the number of participants and the topics increased, a slight reorganisation of the Executive Committee meetings took place in mid of 2020, i.e. the meetings were continued with a reduced number of participants and were focused on more operational topics. In addition, so-called research strategy meetings were introduced with a limited number of participants, i.e. one representative of all involved research institutions, for discussing research programmes (presence and future), common grant applications, beam time distribution, or the like. Last but not least, a seminar on the scientific progress was established with the support of the Medical University of Vienna and the TU Wien for the academic year 2020/2021.

From the beginning of 2019 until mid of 2021, in total 28 Executive Committee meetings took place, which are well documented with respective agendas and memos for each meeting. Due to the Covid-19 situation in 2020 and 2021, the possibility for personal meetings was rather limited. Anyhow, most meetings were organised as online events. In preparation for the Research Strategy 2022 – 2024, twelve dedicated meetings were organised with the participation of all involved research organisations. Starting with October 2020, a new biweekly seminar on "Non-Clinical Research at MedAustron" was introduced for all coworkers and students of the non-clinical research projects. It was organised by the TU Wien and the Medical University of Vienna and was already registered as teaching activity at the Medical University of Vienna and will be listed in the curricula at the TU Wien as well. This non-clinical research seminar was conducted as online meeting and was very well received, which was reflected by the average attendance of more than 20 participants (MSc and PhD students and postdocs).

For the research period from 2019 until 2021, a new structure for an extended Advisory Board was contemplated, i.e. a more direct involvement of the Medical University of Vienna and the TU Wien with their various experts in the field. In the beginning of 2019, this approach was further discussed with all involved stakeholders and finally resulted in an extension of the Advisory Board with two new members, i.e. the Vice Rectors responsible for Research and Innovation form the Medical University of Vienna (Michaela Fritz) and the TU Wien (Johannes Fröhlich), besides Gerald Badurek and the chairman Richard Pötter. One Advisory Board meeting per year took place in December 2019 and December 2020, respectively. For each meeting, a progress report and a proposal for beam time was prepared and presented to the members of the Advisory Board. The required beam time was then recommended by the Advisory Board for the specific projects. After the meetings, dedicated reports were written, which were then forwarded to the representatives of the Austrian Federal Ministry of Education, Science and Research.

According to the contract between MedAustron and the Austrian Federal Ministry of Education, Science and Research, the available beam time per month increased from five shifts in 2019 to eight shifts in 2020 and 2021. Due to the additional beam time provided already in 2018, the total number of shifts for the years from 2019 until 2021 was decreased by four shifts, i.e. $5 \times 12 + 8 \times 12 + 8 \times 12 - 4 = 248$. Given the demanding Covid-19 situation in the last years, the scheduling of the beam time had to be adjusted based on a short-term and flexible mode, which resulted in 61, 88, and 99 (planned) performed shifts for the years 2019, 2020, and 2021 (planned). These figures sum up to an intended total number of $61 + 88 + 99 = 248$ shifts, which is in accordance with the required number of shifts for

this period determined in the above mentioned contract. The actual used beam time by the different non-clinical research projects is summarised in Table 1.1 for the years 2019, 2020, and 2021 (planned). Figure 1.1 shows the distribution of the non-clinical research shifts to the different projects. All these shifts were presented to and recommended by the Advisory Board.

The project on "Pre-Clinical Animal Research" was closed by end of 2020 due to the long-lasting selection process for the assistant professor in Molecular Radiobiology at the Medical University of Vienna. Nevertheless, the preparation work for this project was concentrated more on technical irradiation aspects and the associated medical physics issues for the establishment of an appropriate treatment planning and animal irradiation workflow with the already available hardware and software. Since these activities were more related to commissioning, the required beam time was included in project "Commissioning and Quality Assurance Procedures" for 2021. As mentioned above, the project on "Accelerator Physics" started only in 2020, see Section 1.7. Thus, no beam time was used in 2019. In 2020, the Austrian Space Forum, i.e. ÖWF, used in cooperation with the TU Wien and the Medical University of Vienna two shifts in order to test candidate textiles of a lunar spacesuit for their resistance to high-energy particle radiation and their shielding capacities. The project was performed in the framework of an ongoing Master's project at the TU Wien. Additionally, together with partners from the ÖWF, a publication of the achieved results has been prepared and is currently being extended for specific details of the spacesuit project "Planetary Exploration Textiles (PExTex)" by experts from the European Space Agency.

Consultation with External Experts

In 2019, the Advisory Board recommended to involve experts in the evaluation of the progress and scientific outcome of each project, as defined in the previous Research Strategy covering the period 2019 – 2021. Furthermore, it was agreed that such external peer review could be helpful to steer or redirect research and to prepare the next Research Strategy.

Based on this recommendation and the initiative of Albert Hirtl, who aimed for a workshop with external experts concerning the project on "Imaging with Ion Beams", individual meetings were planned for each project in 2020. Originally, it was envisaged to hold these review meetings during international scientific conferences and workshops in which the experts participate. Unfortunately, due to the Covid-19 pandemic this idea had to be abandoned and review meetings with external experts were held as virtual meetings instead. The respective project description as outlined in the Research Strategy 2019 – 2021 and recent publications were forwarded to the experts in advance and during the virtual meetings, the groups presented current research activities as well as planned ones for 2021. The external experts summarised the conclusions from the review meetings and their recommendations for the respective projects in a written report. These reports were forwarded to the Advisory Board and the major facts were presented during the Advisory Board meeting in December 2020.

All projects except P 2 "Interfraction Adaptive Radiation Therapy" and P 6 "Pre-Clinical Animal Research" could be evaluated thanks to the commitment of the following experts:

- Markus Stock, MedAustron, review of P 0 "Commissioning and Quality Assurance Procedures";
- Antje-Christin Knopf, University of Groningen, review of P 1 "Intrafraction Adaptive Radiation Therapy";

project	2019	2020	2021	total
Commissioning and Quality Assurance Procedures	21	15	18.5	54.5
Intrafraction Adaptive Radiation Therapy	7	9	15	31
Interfraction Adaptive Radiation Therapy	0	3	3	6
Imaging with Ion Beams	8	13	18	39
Magnetic Resonance Guided Particle Therapy	8	10	10	28
Energy Transfer Mechanisms and Applications in Biology and Physics	17	23	23.5	63.5
Pre-Clinical Animal Research	0	4	0	4
Accelerator Physics	0	9	11	20
Austrian Space Forum, Österreichisches Weltraum Forum (ÖWF)	0	2	0	2
total	**61**	**88**	**99**	**248**

Table 1.1: Summary of the performed non-clinical research shifts per topic or project for the last three years, i.e. 2019, 2020, and 2021 (forecast).

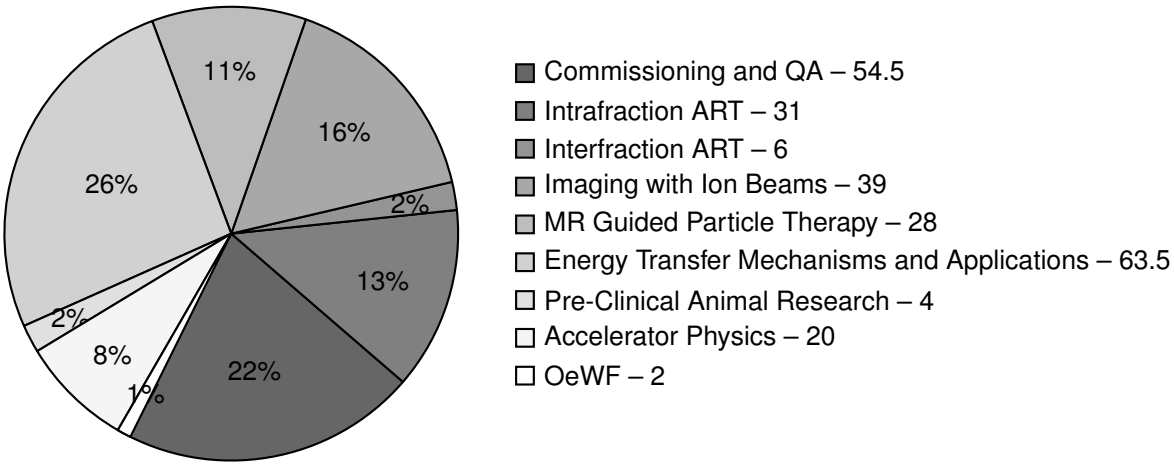

Figure 1.1: Distribution of the 248 non-clinical research shifts to the different research projects in the research period from 2019 until 2021 as described in the text.

- Reinhard Schulte (Loma Linda University) and George Dedes (Ludwig-Maximilians-Universität München), review of P 3 “Imaging with Ion Beams”;
- Björn Poppe, University of Oldenburg, review of P 4 “Magnetic Resonance Guided Particle Therapy”;
- Naruhiro Matsufuji, National Institute of Radiological Sciences, review of physic research in P 5 “Energy Transfer Mechanisms and Applications in Biology and Physics”;
- Nils-Petter Rudqvist, MD Anderson, review of biology research in P 5 “Energy Transfer Mechanisms and Applications in Biology and Physics”;
- Michael Benedikt, CERN, review P 7 “Accelerator Physics”.

Overall, this review cycle by external experts which was conducted as pilot project in 2020 was well received by both the non-clinical research teams and the Advisory Board. In light of the preparation for the new research strategy 2022 – 2024, no such external expert review was conducted in 2021, but is planned for 2022.

Accepted Grant Applications

1. FWF: “Proton and Carbon Ion Response Studies for Chondrosarcoma” was submitted by the Medical University of Graz and the Medical University of Vienna in September 2018. The project duration is from October 2019 until September 2022.
2. FFG BRIDGE Programme: “Image Guided Particle Therapy of Uveal Melanoma with a Multi-Port Beam Arrangement (IMPACTOR)” was submitted by MedAustron and the Medical University of Vienna in April 2019. The project duration is from October 2019 until March 2022.
3. FFG Industrienahe Dissertationen: “Bildgebung mit hochenergetischen Protonenstrahlen für eine verbesserte Krebsbehandlung” was submitted by the HEPHY in May 2019. The project duration is from December 2019 until November 2022.
4. EC Marie Curie ITN: “Real-Time Adaptive Particle Therapy of Cancer (RAPTOR)” was co-submitted by the Medical University of Vienna (beneficiary) and MedAustron (partner) in January 2020. The project duration is from March 2021 until February 2025.
5. NÖ doctoral grant programme: “Fully Integrated 4D Dose Calculation and Beam Delivery Optimisation in Particle Therapy” was submitted by MedAustron in February 2020. The funded PhD student started at MedAustron in 2020. The project duration is from April 2020 until March 2023.
6. EC Horizon 2020 – Integrating and Opening Research Infrastructures of European Interest: “PRoduction of hIgh purity iSotopes by MAss Separation for Medical APplication (PRISMAS-MAP)” was co-submitted by MedAustron in May 2020. The project duration is from April 2021 until March 2025.
7. EC Horizon 2020 – Integrating and Opening Research Infrastructures of European Interest: “Heavy Ion Therapy Research Integration *plus* (HITRI*plus*)” was co-submitted by MedAustron in May 2020. The project duration is from April 2021 until March 2025.

8. NFB Grant: "Advanced Molecular Imaging for Personalized High-Precision Radiation Therapy" was submitted by the University of Applied Sciences Wiener Neustadt in March 2020. The project duration is from March 2021 until March 2026.

9. FFG BRIDGE Programme: "Hi-Precision Beam Position and Intensity Monitor for Accurate Cancer Treatment with Ions (HiBPM)" was submitted by the HEPHY and MedAustron in September 2020. The project duration is from April 2021 until March 2024.

10. FWF doc.funds.connect Programme: "Pre-clinicAl Ion beam Research (PAIR)" was submitted by the Medical University of Vienna, the University of Applied Sciences Wiener Neustadt and the University of Veterinary Medicine Vienna in November 2021. The project duration is four years and will start in 2022.

2 Motivation and Vision for 2022 – 2024

During the first research period 2015 – 2018 both, the research organisation and the envisaged research topics, were following a pillar like structure based on professorships at the Medical University of Vienna and the TU Wien. After gaining research experience and consolidating the research groups from the different universities, a more integrative and interdisciplinary approach across the user groups from the various institutions including MedAustron was implemented during the second research period 2019 – 2021. In this more horizontally organised setting, the following main research and development activities were pursued:

- Commissioning and Quality Assurance Procedures
- Intrafraction Adaptive Radiation Therapy
- Interfraction Adaptive Radiation Therapy
- Imaging with Ion Beams
- Magnetic Resonance Guided Particle Therapy
- Energy Transfer Mechanisms and Applications in Biology and Physics
- Pre-Clinical Animal Research
- Accelerator Physics

Several of the above-mentioned research projects very already initiated during the first research period and/or supported by research grants and thus continued while others were established only in the second research period.

From an organisational point of view and in light of the given person power situation, it was agreed by all participating user groups that the granularity of the research projects and research organisation needs to be reduced. In addition, it was agreed that for visibility reasons towards the national and international research community, the research topics should be condensed. For all these reasons, a research project re-organisation was discussed and the following four new research pillars were established for the third research period 2022 – 2024:

- Applied Particle and Medical Physics

- Biophysics and Molecular Radiobiology
- Technological Innovations and Clinical Implementation
- Accelerator Physics

In a similar manner as in the second research period, two project coordinators were assigned to each research pillar, which is then further broken down into work packages. The already well established and fruitful collaboration across intuitions is reflected in this research strategy and another successful research period is to be expected.

2.1 Exploratory Studies and Emerging Topics

To accommodate flexibility to test new research ideas or to facilitate a defined way to request more beam time on short notice if required by a project, a small amount of dedicated beam time will be reserved in each research field. The reasons are manifold. Firstly, during the revision process of manuscripts submitted to peer reviewed journals it was requested to perform additional experiments to corroborate the reported results. So far, such additional work could be only performed by reducing other planned research and development activities. Secondly, it was experienced that additional research topics became very relevant during a three years period. In order to prepare for such upcoming topics and to be able to perform respective pilot studies, a certain amount of spare beam time on top of the one needed for ongoing projects is necessary. Thirdly, the community of potential beam time users in Austria is slowly growing. As these groups have no practical experience and thus detailed research plans are lacking, it is difficult to prospectively predict their real needs with regard to beam time. For all these reasons, a more flexible allocation of beam time is needed and therefore it was decided to include a lump sum of available beam time for "Exploratory Studies and Emerging Topics" in the current research strategy. Beam time from that contingent should be discussed within the respective research fields and formally requested in the regular Executive Committee meetings. Such a low-threshold access to beam time will certainly allow for introducing fresh ideas into established research projects and will thus contribute to the success of non-clinical research at MedAustron in general.

Aerospace Irradiation Studies

Recently, a consortium of the University of Applied Sciences Wiener Neustadt, FOTEC (the research subsidiary of the University of Applied Sciences Wiener Neustadt), and the Seibersdorf Laboratories expressed their interest to test electronics devices used for space applications, e.g. for satellites, at MedAustron. So-called single event effects are a crucial concern in developing and qualifying technologies for aeronautic and space applications, requiring appropriate irradiation tests with, for example, 200 MeV protons during the development process. In order to establish the necessary expertise on-site to carry out such measurements, it is intended to use the first beam time from the contingent of "Exploratory Studies and Emerging Topics". Depending on further interest in this topic, it is planned to formulate dedicated research projects that will require beam time in the next years.

2.2 Executive Summary of Proposed Research Fields

Applied Particle and Medical Physics

With the successful testing of an ion-imaging demonstrator based on available technology for particle tracking and residual energy measurement in the last research period, a new international actively pursued research topic was established at the TU Wien and the HEPHY. The initiated activities will be continued in the upcoming research period by working on significant upgrades of that demonstrator system based on detectors with high time resolution allowing for time-of-flight residual energy determination, thus introducing a new paradigm to the field. Closely related to the developments in ion imaging, novel sensors for particle physics in general and for an upgraded beam monitoring system at MedAustron will be tested.

Dose determination for high-precision particle therapy is an important medical physics research topic at MedAustron. Time-resolved dosimetry, dosimetry in the presence of magnetic fields, and device characterisation are the main pillars that will be investigated in the upcoming research period, accompanied by Monte Carlo simulations. Methodologies for 4D dosimetry will be developed, as well as correction factors and protocols for dosimetry in magnetic fields. Furthermore, it is envisaged to establish MedAustron as a centre of competence for detector development and testing.

Another branch of dosimetry, microdosimetry, was started in the last research period in the framework of two PhD projects. The aim is to achieve a coherent and standardised physical characterisation of the irradiation field, considering radiobiological models as well as studies on cell lines. Using solid-state and gas detectors, energy spectra are determined on cell-sized volumes and can thus be correlated to biological effectiveness.

All mentioned activities rely on the support from well-trained students. Therefore, a lab course introducing interested students to scientific methods used in experimental particle physics and medical physics was established at the TU Wien.

Biophysics and Molecular Radiobiology

Linking radiobiology research with high-quality physics data is a prerequisite for investigating biological effects of particle radiation. Robust dosimetric procedures for cell experiments in the frame of radiobiology research has been established within the last research period through a strong and fruitful collaboration between biology and medical radiation physics. Collaborative efforts will be continued and extended to include in vivo research using mouse models coupled with multimodal imaging for the correlation and visualisation of dose and LET effects. This will require intensive efforts on the establishment of high-precision irradiation and imaging workflows for photons as well as charged particles.

Our research will continue to focus on radiation-induced anti-tumour immunogenicity, tumour heterogeneity and tumour microenvironment. We will combine moderate immunogenic alterations in human pancreatic cancer in vitro models, with comparable effects between protons and X-rays. Next, we will focus on anti-tumour immunogenicity following proton and carbon ion exposure and combinatorial approaches including DNA damage response (DDR) inhibitors.

Furthermore, in vivo research will be deployed to comprehensively address the crosstalk between tumour cells and tumour microenvironment, furthering the development of personalised ion-beam therapy approaches. We will apply morphological and functional imaging approaches comprising micro-PET, micro-SPECT, micro-CT, and UHF strength micro-MRI

approaches to determine in vivo treatment response. Through interdisciplinary collaborations among physicists, biologists, biochemists and imaging specialists, we will form a hub for modern radiobiology and biophysics research.

Technological Innovations and Clinical Implementation

Based on the successful establishment of a 4D dose tracking research framework, artifical intelligence (AI) based image processing, sCT generation from MR data sets and the start of a comprehensive clinical data collection in the context of retrospective data analysis will be continued and extended in close collaboration with the EBG MedAustron.

As extension of the MR-only workflow developed recently, the investigation of novel MR sequences for sCT creation as well as treatment planning studies will be initiated, aiming to pursue the next steps towards online-MR guided particle therapy.

The retrospective analysis of clinical data for moving targets serves as decision criteria for respective motion mitigation techniques, which will be validated in well-designed end-to-end tests also in the frame of the recently started EC funded project RAPTOR. Next steps will focus on automation and will encompass dose accumulation for intrafractional and interfractional moving targets, as well as clinical outcome analysis based on anatomical combined with functional imaging and treatment data.

The project to treat of uveal melanoma patients in lying position was launched during the last research period and will be finalised in the coming years. Hardware and software developments conducted in the frame of this FFG funded project are expected to be finished in 2022. This will be followed by the clinical implementation and validation of this innovative treatment method which is envisaged to be completed in this research period.

Research and development conducted in this pillar "Technological Innovations and Clinical Implementation" relies on excellent calibrated equipment and constant beam characteristics. To all intents and purposes, this holds for all other non-clinical research pillars, too. Therefore, regular QA measurements based on procedures established over the last years, will be performed and extended where necessary, throughout the entire research period.

Accelerator Physics

The commissioning of helium ions for the different accelerator parts at MedAustron started already in the last year. Within the next three years, the activities will concentrate on the linear accelerator (LINAC), the medium-energy beam transport (MEBT), and the synchrotron commissioning in order to provide a proper helium-ion beam for the HEBT and finally in IR 1 in the year 2025, i.e. in the beginning of the upcoming research period. This project requires a considerable effort in both, the expertise of the accelerator team with the support of competent students and beam time.

Besides the currently used third order slow-resonant extraction technique in the MedAustron synchrotron, there are alternatives providing potential new features for the overall performance of the facility. Research on COSE, radio frequency (RF) noise driven extraction, and radio frequency knock-out (RFKO) will be performed by means of simulation toolkits and machine experiments using existing systems and, where possible, flexible prototyping hardware. With the establishment of these new techniques, an improved beam quality, multi-energy extraction (MEE), and a dynamic intensity control will become accessible.

A dedicated work package provides the opportunity to follow and explore novel developments in accelerator physics for ion-beam therapy. Topics such as ultra-high dose rate (FLASH) radiotherapy with ion beams or mixed beam arrangements during the entire acceleration process will be addressed to the next generation of accelerator physicists in terms of project and diploma theses.

2.3 Summary of Estimated Beam Time Request

The requested beam time for 2022 is listed in the respective project descriptions below for each work package, whereas for the years 2023 and 2024, the requested beam time will be justified on a yearly basis and justified in details in the regular documents for the Advisory Board meetings. At this stage, as current best estimates, a continuation of beam time per research field is displayed in this document, taking into account the annual increase in beam time. This way a more reliable beam time distribution can be established on the basis of project requirements in the respective year. The available beam time for non-clinical research at MedAustron is detailed in Table 2.1.

	2022	**2023**	**2024**
Applied Particle and Medical Physics	**46**	**45**	**50**
Particle Imaging	6	6	6
Instrumentation	9	9	9
Dose Determination	15	14	18
Microdosimetry	15	15	16
Laboratory Course	1	1	1
Biophysics and Molecular Radiobiology	**17**	**19**	**21**
Pre-Clinical Animal Research	7	7	7
Translational Radiobiology	10	12	14
Technological Innovations and Clinical Implementation	**10**	**12**	**11**
Clinical Implementation and Validation	3	5	4
MRI in Particle Therapy	2	3	3
Quality Assurance	5	4	4
Accelerator Physics	**42**	**46**	**51**
Implementation of Helium Ions	25	28	23
Extraction Mechanisms	15	14	24
Novel Developments	2	4	4
Exploratory Studies and Emerging Topics	**5**	**10**	**11**
Total	**120**	**132**	**144**

Table 2.1: Estimated beam time request for the different work packages in the upcoming three years.

Field of Research

Applied Particle and Medical Physics

Coordinators:	Albert Hirtl and Peter Kuess
WP coordinators:	WP 1 – Albert Hirtl WP 2 – Thomas Bergauer WP 3 – Peter Kuess WP 4 – Giulio Magrin WP 5 – Albert Hirtl
Co-operators:	Medical University of Vienna, Technische Universität (TU) Wien, EBG MedAustron

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Required Beam Time **45**

This research field serves to establish a sharpened research profile in areas of expertise of the involved groups and to continue successful projects started in the previous research periods under a comprehensive and yet flexible umbrella with regard to the extension to new topics in the future.

With "Particle Imaging", a new and internationally actively pursued topic was started at the TU Wien and the Institute of High Energy Physics (HEPHY) in the last research period and led to the implementation of an ion-imaging demonstrator based on available detectors and was tested with protons at MedAustron. The success was enabled by the long-standing expertise in detector technology and infrastructure at the HEPHY and will be continued by starting a development effort towards a clinically useful system based on fast sensor technology such as low gain avalanche detectors (LGADs) for time-of-flight (TOF) energy measurement. The overall long-term goal of this effort is the development and in-depth evaluation of TOF-based ion computed tomography (CT). This includes the implementation of a technology demonstrator, an optimised and scalable system design benchmarked in Monte Carlo simulation, and optimised image reconstruction algorithms.

In synergy with the work package on "Particle Imaging", new instrumentation for nuclear and particle physics experiments and upgraded beam monitoring for the MedAustron beam line will be tested. Conclusions from these tests will certainly also have an impact on the particle imaging work package and will further strengthen the collaboration among the involved partners.

Already in the previous research periods, dose determination in the context of high precision ion-beam therapy proofed an important medical physics research topic. The successfully conducted dosimetric experiments with the research magnet in irradiation room 1 (IR 1) in the last research period will be continued and intensified for protons and carbon ions. Furthermore, the characterisation of novel detectors with protons and carbon ions started in the last research period within a cooperation with industrial partners. This cooperation is planned to be continued and intensified. Dosimetric work is always accompanied by Monte Carlo simulations, playing an important role in this research topic.

Microdosimetry investigations started in the previous programme in parallel to the conventional dosimetry research. The goal is to achieve a coherent and standardised physical characterisation of the irradiation field. Henceforth, radiobiological models can complement comprehensive physics characterisation into cellular studies and clinical practice. Solid-state and gas detectors are used to collect spectra of the energy imparted from proton and carbon-ion beams to cell-like volumes. These spectra provide a reliable and repeatable characterisation of the irradiation fields, which can be directly correlated to the biological effectiveness. In the framework of two complementary PhD programmes, young researchers are investigating the challenges of implementing the different novel detectors in high-intensity ion beams, the peculiar characteristics of the microdosimetric spectra obtained, and the implementations of the results in radiobiological and clinical researches.

Well educated students in technology and approaches used in experimental particle physics and medical physics are an integral part and important support for the successful implementation of the pursued research topics elaborated in this document. Therefore, to give a regular update on the research conducted at the research beam line of the MedAustron facility and to

discuss new ideas, the seminar series *Non-clinical Research at MedAustron* has been initiated in the winter term 2020. This seminar was well received and has thus become part of the curricula at the TU Wien and the Medical University of Vienna with the title *Seminar on Medical Radiation Physics and Ion-Beam Therapy.* It is organised by Dietmar Georg, Albert Hirtl and Barbara Knäusl. Additionally, the laboratory course *Experiments at the MedAustron Particle Accelerator – Applied Particle Physics and Medical Physics* and the dedicated lecture *Introduction to Medical Physics Aspects of Ion-Beam Therapy* were added to the curricula at the TU Wien.

WP 1 Particle Imaging

Objectives

- availability of a functional ion CT demonstrator based on TOF residual energy measurement

WP 1.1 Development of Time-of-Flight Ion Computed Tomography Demonstrator

In this work package, different low gain avalanche detector (LGAD) designs will be investigated with respect to their usability in an ion-imaging apparatus. First, small samples of LGADs will be obtained via academic partners from existing productions. The samples will then be tested and characterised using laboratory equipment, picosecond lasers and with protons in the research beam line of the MedAustron facility. The experimental effort will be accompanied by finite element semiconductor simulations (Synopsys TCAD) as well as detailed Monte-Carlo detector simulations (Allpix Squared, Weightfield) whenever required. This will also entail studies with regard to system integration aspects such as cross talk, achievable data rates, signal duration, sensor occupancy and pile-up. Once the LGAD technology has been identified based on the tests outlined above, a dedicated production run will be initiated at a silicon foundry. Funding for this production is available via the TU Wien.

For the ion CT demonstrator, a multi-channel readout board with discrete analogue components will be designed and verified by electronic circuit simulation. Digitisation will be achieved by field programmable gate array (FPGA)-based time-to-digital converters (TDCs). In parallel, a scalable module concept based on an integrated application-specific integrated circuit (ASIC) solution will be developed with an ion CT system of clinical dimension in mind. This work will orient itself at on-going ASIC developments in WP 2 “Instrumentation” as well as existing design work for modules such as for the Belle-II Silicon Vertex Detector (SVD). Finally, the various stages of the proposed demonstrator will be tested experimentally. Besides conducting small-scale lab tests at the TU Wien and the HEPHY, most data will be acquired at MedAustron.

WP 1.2 Image Reconstruction and Monte Carlo Simulation

The main focus in this work package will be to adapt and optimise existing reconstruction algorithms to the new TOF paradigm. This will be done with Monte Carlo (MC) simulation using the open source particle transport toolkit Geant4 or, more conveniently, its macro

implementation Geant4 application for tomographic emission (GATE). Topics to be studied include calibration aspects and track reconstruction with precise timing at high-track densities. Several methods to improve the spatial resolution achieved via sophisticated path estimation (e.g. cubic spline) will be studied. To make the used reconstruction framework as flexible as possible and to allow its use also for other imaging modalities (e.g. fluence loss imaging or scattering tomography) and other ion species (e.g. helium ions), adjustments in the used reconstruction software are planned. The future availability of helium ions at MedAustron and their promising potential in ion imaging is particularly interesting and will be investigated in detail and optimised with MC simulations and measured data, if available.

Main Involved Personnel and Timeline

The work package will be coordinated by Albert Hirtl (TU Wien) and Thomas Bergauer (HEPHY), reflecting the close collaboration of the two institutions. The work package implementation of the intended activities will be supported by electrical engineers and technicians from HEPHY and by postdocs and PhD students. Currently, two PhD students are funded and will continue to work on their respective tasks until beginning 2022 and the end of 2022. The experience from the last years showed that on average two diploma students and several project students for smaller tasks are expected to contribute per year. These undergraduate students get the opportunity to actively participate in the beam times and receive hands on training in an experimental physics programme in an accelerator environment.

The experimental effort and the work on image reconstruction and MC simulation will be done in parallel, whereby MC simulations will support experimental decisions as needed. Following tests of various technology candidates, the most suitable technology will be chosen and a production run at a silicon foundry will be initiated. Once delivered, the detectors will be thoroughly characterised and eventually the demonstrator system will be assembled and its functionality will be tested at HEPHY. For obtaining data with the full demonstrator, the set-up will be transferred to MedAustron and tested in scheduled beam times. The final objective will be to acquire images of a suitable phantom utilising the demonstrator's field-of-view and TOF residual energy measurement. The images will be reconstructed using software developed in WP 1.2 "Image Reconstruction and Monte Carlo Simulation". Further developments in image reconstruction are planned to be supported by experts at CREATIS, Lyon (S. Rit), with whom a collaboration has been started in the previous research period. In Figure 1.1 an estimate for the project timeline is given.

WP 2 Instrumentation

Objectives

- availability of a flexible set-up for detector tests
- availability of a detector system for measuring particle beam profiles and positions in a wide beam intensity range in the MedAustron beam line

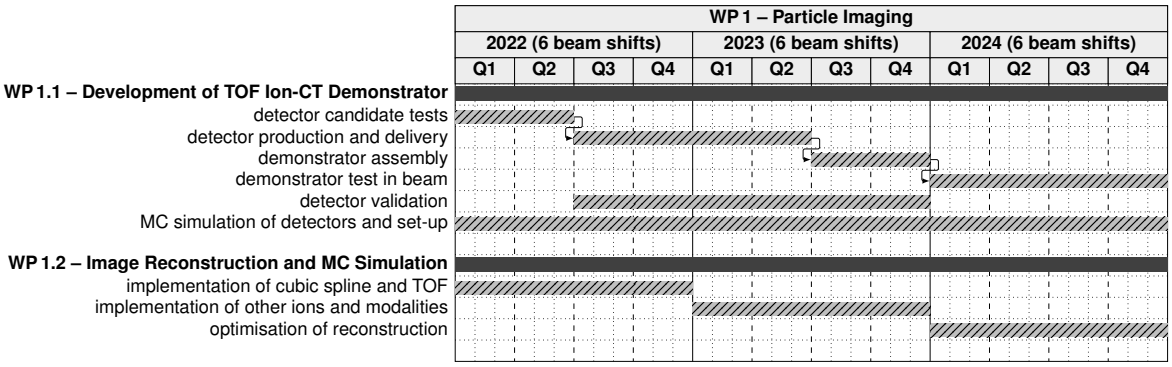

Figure 1.1: Timeline for WP 1 "Particle Imaging".

WP 2.1 Novel Detectors for Particle and Nuclear Physics Experiments

Particle physics experiments rely on the development of innovative detector technologies and concepts. Prior to be used in an experiment, these detectors need thorough testing and characterisation in suitable particle beams, ideally with minimum ionising particles (MIPs). At MedAustron, 800 MeV protons, which is close to the MIP energy for silicon, are available, thus offering a good testing environment for semiconductor detectors.

Given the experience from the last research period, where new detector technologies interesting for research pursued at the HEPHY or the TU Wien emerged regularly, the requirement of a dedicated work package and beam time for their testing occurred. Therefore, the aim of this work package is to enable systematic testing of new cutting edge detector technologies (e.g. monolithic HV-CMOS detectors or detectors with high time resolution) and their electronics in the MedAustron beam line with an adequate amount of beam time. Besides, the development work for a demonstrator system in WP 1 "Particle Imaging" will certainly also profit from new technologies identified in the scope of this work package.

To facilitate a convenient possibility for detector testing, a flexible test-set-up will be developed where new detectors can be operated easily.

WP 2.2 Hi-Precision Beam Position and Intensity Monitor

The aim of this work package is to experimentally support the funded Bridge-1 Austrian Research Promotion Agency (FFG) project *Hi-Precision Beam Position and Intensity Monitor (HiBPM)*. Within this project, a new detector system enabling the measurement of the beam profile and position in a wide intensity range will be developed, which could eventually replace the currently existing beam monitoring system at MedAustron based on scintillating fibres. The development of an ion-imaging demonstrator in the previous research period required the availability of reduced particle flux at MedAustron. Therefore, three intensity reduction modes have been established, which are suitable for the existing ion-imaging demonstrator and also research projects pursued by other groups (e.g. microdosimetry). As a result, the beam intensity for these modes became too low to perform regular quality assurance (QA) measurements of the beam profile with the standard equipment in the MedAustron beam lines. This applies especially for the 30 installed beam position monitors. For this purpose, radiation-hard solid-state sensors will be studied. Several sensor candidates, based on silicon

carbide (SiC) as base material, will be tested. This so-called "wide band-gap" material is relatively new for particle detectors and became a realistic option due to the increasing demand of the chip industries for its application in power devices. The studies include testing detectors of different sizes and thicknesses to evaluate the technologies limits, as well as evaluating sensors with intrinsic gain. Additionally, different readout-electronics optimised for the SiC sensors will be tested, ranging from discrete single-channel electronics up to ASIC-based readout electronics. Both, sensor and electronics prototypes will be investigated with particles from radioactive sources at the HEPHY and with particle beams at MedAustron, supported by MC simulations (Weightfield2, GEANT4, Allpix2, TCAD).

Main Involved Personnel and Timeline

The work planned in this work package will be coordinated by Thomas Bergauer (HEPHY). The implementation of sub-work package WP 2.1 "Novel Detectors" will be supported by PhD students and undergraduate students from ongoing projects at the TU Wien and the HEPHY aiming at testing detectors for particle physics research and by electrical engineers and technicians from the HEPHY. Sub-work package WP 2.2 "Hi-Precision Beam Position and Intensity Monitor" is fully funded by a common grant submitted to the FFG by the HEPHY and MedAustron and includes one postdoc and two PhD students for three years at the HEPHY. Although the project started officially in April 2021, hiring of the first personnel took until September. Therefore, the project duration will be extended accordingly, which is a well established procedure at the FFG.

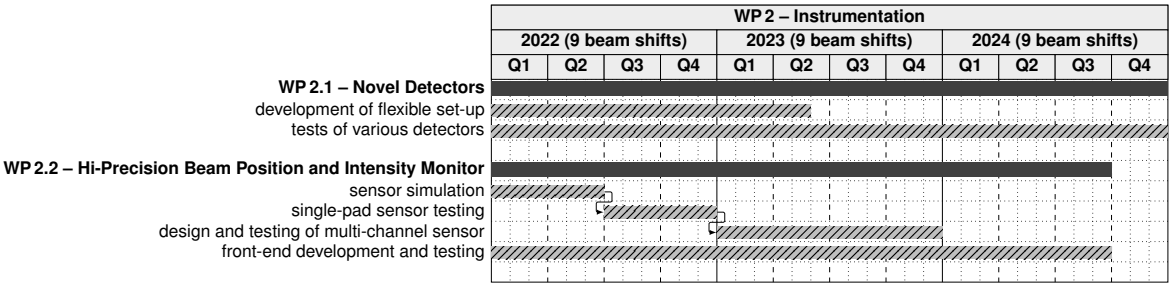

Figure 2.1: Timeline for WP 2 "Instrumentation".

WP 3 Dose Determination

Objectives

- develop methodologies for 4D dosimetry
- determine detector correction factors and protocols for dosimetry in magnetic fields
- establish MedAustron as a centre of competence for dosimeter testing

WP 3.1 Time Resolved Dosimetry

Time resolved dosimetry encompasses all dosimetric endeavours related to changes in time, both within a treatment fraction (intrafraction) as well as between different treatment fractions (interfraction). Time resolved dosimetry provides significant insights in the four-dimensional pencil-beam scanning (PBS) delivery dynamics, the resulting accumulated dose, and potential sources of errors and uncertainties.

Intrafraction

The major aim of this work package is to identify motion amplitudes, which influence the dose delivery outcome, and to optimise delivery dynamics for moving targets. This will be investigated for all available particle species and with respect to possible motion mitigation techniques for the technical infrastructure at MedAustron. These measurements are used for the validation of the computational dose tracking framework for all particle species. The well tested breathing phantom advanced radiation dosimetry system (ARDOS) and the existing time resolved dosimetry framework builds the basis for this endeavour and will be extended with new phantoms and dosimetric devices. More details about the computational four-dimensional dose tracking framework and the retrospective analysis of patient data are described in WP 1.1 "Outcome Modelling and Automatisation" of the Research Field "Technological Innovations and Clinical Implementation".

Interfraction

Another part of this work package is to extend currently established dosimetric procedures for static end-to-end tests towards real-time adaptive proton therapy components. This will be performed within an European Commission (EC) funded project, i.e. "Real Time Adaptive Particle Therapy of Cancer (RAPTOR)". The development and experimental validation of a novel 4D anthropomorphic phantom is the core activity of this project and will be performed in close cooperation with the University Medical Centre Gronigen (UMCG). Aim is to develop and validate straightforward dosimetric workflows that are universally applicable.

WP 3.2 Dosimetry in Magnetic Fields

The overall goal of magnetic resonance guided particle therapy (MRgPT) is to be able to perform on-line adaptive treatments. Dosimetry in magnetic fields plays here an important role regarding beam characteristics, biological experiments, treatment planning system (TPS) validation, and QA. Currently, there are no dosimetric protocols and correction factors available to perform dosimetry in magnetic fields. The magnetic fields from the magnetic resonance imaging (MRI), homogeneous as well as fringe fields, induce changes on the beam paths during the treatment, leading to geometric and consequently, dosimetric changes. The influence of the magnetic field on dosimetric measurements can be expected to be substantial compared to the overall uncertainty of 1 % aimed for in reference dosimetry for particle beam therapy. It is thus important that the mechanisms that contribute to these alterations are meticulously studied and quantified. Furthermore, the magnetic fields from the magnetic resonance (MR) core will affect currently employed dose verification methods, due to the impact of magnetic fields not only on the dose delivery, but on the response of standard clinical detectors. The effect on dose distributions needs to be quantified and the changes in charged particle fluence

may also require modifications to the cavity theories that are used to convert energy deposition in a detector volume to dose to the medium. An important effect in this respect can be expected from the change of the charged particle fluence perturbations that are the results of the presence of the detector. But also, the operational principle of detectors themselves can be affected.

The most common instruments to determine dosimetric quantities are based on the measurement of ionisation in a material which is not radiologically equivalent to water or tissue (e.g. air in an ionisation chamber and silicon or diamond in diodes). Because of the action of the magnetic field on the paths of all charged particles, these corrections for fluence perturbations are altered. For high-energy photon beams, it has been shown that the change of the secondary electron fluence can result in a change of response of a typical reference ionisation chamber by up to 11 % depending on the detector orientation and for diamond dosimeters, the effect can even be larger than 20 %. Results from the previous research period showed smaller, but still significant effects, which implies the need for further measurements. In general, the reasons for these changes are unidentified.

Furthermore, ion recombination effects, both initial and volume, in ionisation chambers will be investigated with respect to the presence of a magnetic field. Due to the different mechanisms by which the influence of the magnetic field manifests itself, it is important to develop a way of separating these two effects experimentally so that theoretical models developed within this project can be validated. A cooperation with the University of Oldenburg (Germany) was initiated to better characterise and model detector response in magnetic fields. In addition, a cooperation with the Centre for Radiation Research in Oncology (OncoRay) in Germany was already initiated in 2021 aiming to establish and improve a dosimetry workflow and QA process for MR guided proton therapy.

WP 3.3 Device Characterisations

Commercially available dosimeters (e.g. ionisation chambers) are typically tested and calibrated in ^{60}Co beams, available in standard laboratories. The transfer to particle beams is often challenging for the vendors of such equipment and thus equipment is often released before extensive tests with particle beams were performed. The MedAustron infrastructure, enabling proton and carbon-ion beams, enables options for detailed device characterisations. In addition, such characterisation is also of great interest for designing novel dosimetric equipment and QA procedures. Thus, the aim is to establish MedAustron as a centre of competence for dosimeter testing.

MedAustron has a very good cooperation with Ion Beam Applications (IBA). Within this cooperation, several novel dosimetric equipment will be investigated in the clinics at MedAustron for QA purposes. This equipment includes an upgraded scintillation detector, i.e. Lynx 2, plane-parallel StingRay ionisation chambers, and a Phoenix high-resolution digital detector. Furthermore, an XRV 124 three-dimensional digital camera phantom from Logo Systems will be available for gantry commissioning, which was tested with a focus on proton dosimetry. This implies potential research activities with carbon ions.

Novel dosimetry detectors will be tested and characterised in multiple ways. Two-dimensional measurements, such as spot position and shape measurements, can thus be performed with five different devices, i.e. radiochromic films, Lynx, Lynx 2, Phoenix, and XRV 124. A thorough comparison between these devices for proton and carbon ions and eventually including 800 MeV protons or low flux settings has a high publication potential.

One-dimensional device characterisation remains an important part for absolute dosimetry. Recombination effects, for example, require thorough investigations. These effects are caused by the recombination of positive and negative charges before they can be collected in a device that measures ionisation. They are particle type dependent and thus challenging to determine before novel detectors are available on the market. In ionisation chambers, two main recombination processes can be distinguished. Initial recombination that occurs when charges formed within the same ionising track recombine and volume recombination between charges formed by different tracks. The latter is proportional to the dose rate or dose per pulse.

Main Involved Personnel and Timeline

The work will be coordinated by three postdocs from the Medical University of Vienna. In detail, Barbara Knäusl will coordinate WP 3.1 "Time Resolved Dosimetry", Hermann Fuchs WP 3.2 "Dosimetry in Magnetic Fields", and Peter Kuess WP 3.3 "Device Characterisations". WP 3.1 is supported by two PhD students, one until the end of 2022 (i.e. Franciska Lebbink) and one for the entire research period (i.e. Jacob Brunner). Further support is provided by Monika Clausen (Medical University of Vienna) and Hugo Palmans (EBG MedAustron) as dosimetric specialist. Additionally, undergraduate students will support this research. They will also contribute as shifters during beam time.

In September 2021, an Austrian Science Fund (FWF) grant was submitted to support research and development of WP 3.2 "Dosimetry in Magnetic Fields".

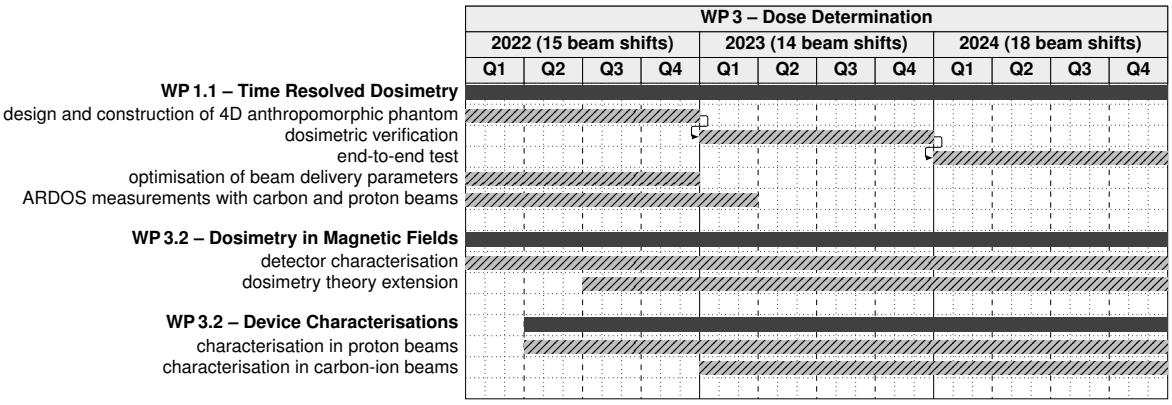

Figure 3.1: Timeline for WP 3 "Dose Determination".

WP 4 Microdosimetry

Objectives

- providing a proposal for the overall uncertainty budget for microdosimetric outcomes (spectra and mean values of lineal energy) making use of laboratory measurements and Geant4 simulations
- characterising two novel microdosimetry prototypes, the telescope E-ΔE diamond detector and the dosimeter-microdosimeter detector
- proposing the gold standard microdosimeters for measurement of proton and carbon-ion beams to complement radiobiological and clinical studies

WP 4.1 Uncertainty Budget for Microdosimetry Analysis

Currently, the comparison of microdosimetric spectra across different institutes and using different technologies is not straightforward due no universally-applied metrological basis in this field and different technical requirements. Based on the experiments carried out in the beam line, a standardisation of microdosimetric procedures and the definition of the appropriate methodology will be possible. Monte Carlo simulations will also be taken into account during this phase. This will allow harmonising the experimental microdosimetric procedures with external users, who could employ microdosimeters in their areas of research and drafting recommendations for the wider community including input for the report on "Microdosimetry" of the International Atomic Energy Agency (IAEA) and the report on "Stochastic Nature of Radiation Interactions: Microdosimetry" of the International Commission on Radiation Units and Measurements (ICRU). Moreover, the above objectives will provide experimental evidence for refining radiobiological models and indicate possible methods for clinical adoption.

The focus of this work package is addressing the impact of the uncertainty on the microdosimetric spectra in a systematic way. This work will progress from the previous research programme, which defined the experimental set-ups and methods to assure reproducibility, and recommended the ad-hoc calibration procedures.

Detector Comparison

It has been shown that currently available detectors are not optimised to perform the microdosimetric characterisation of the full range of ion-beam energies. The study will focus on uncertainties arising from the energy loss straggling, the delta-ray escape, the distribution of particle paths in the detector sensitive volume and experimental contributions (e.g. environmental, electronic influences). The goal is to provide an estimation of the uncertainties, which are not only referring to the mean values of lineal energy $\overline{y}_{\mathrm{F}}$ and $\overline{y}_{\mathrm{D}}$, but also to their spectra. These mean values correspond to the number of events and the delivered dose, respectively. Single, double, and multiple linearisation methods are being investigated to see the impact of each one on the spectra and the $\overline{y}_{\mathrm{F}}$ and $\overline{y}_{\mathrm{D}}$ values, further examination will be done with different stopping-power lookup tables and MC references. The experimental investigations will

continue studying a single irradiation condition using multiple detectors and the conversion of the spectra considering the different materials, shapes, and dimensions.

Correlation to LET Uncertainty

An aspect relevant for both, radiobiological and clinical applications, is the uncertainty correlated to the evaluation of the linear energy transfer (LET) distributions through microdosimetric measurements. High-resolution systematic microdosimetric measurements will be performed in a dense three-dimensional pattern. The track-averaged and dose-averaged LET evaluated through microdosimetry measurements will be compared with values computed via TPSs and MC simulations.

Effects of Counting Statistics

A study of the impact of the uncertainty due to the counting statistics (i.e. the number of detected events used for a single collected spectrum) on microdosimetric spectra and on the frequency-mean and dose-mean lineal energy uncertainties will be carried out. The focus will be, in particular, on the influence of rare nuclear reaction events, which could bring significant variations to the frequency-mean and to the dose-mean lineal energy values, thus a degradation of their uncertainty. Experimental data and specific MC simulations will be compared in relevant conditions to study the impact on the final uncertainty and to provide recommendations on the minimum number of events to be collected in order to stay below a predefined uncertainty.

WP 4.2 Advanced Microdosimetric Prototypes

Dosimeter-Microdosimeter

During the programme, the innovative diamond multipurpose-detector prototype, the so-called dosimeter-microdosimeter (DµD), will be thoroughly studied in an intense programme of irradiation. The DµD should be available for first tests already by the end of 2021, in the framework of the previous research programme. This detector features, in the same substrate, two concentric sensitive volumes, one providing the dose and the other the microdosimetric spectrum. An advantage of the synchronised measurement is that the depth-dose profile obtained with the dosimeter improves the accuracy of the depth position of the microdosimeter, approximately down to a few tens of micrometers. Such accuracy would be impossible to reach using other experimental set-ups. Additionally, the dosimetric information can be applied to compensate the inaccuracies of the mean values of lineal energy, $\overline{y}_{\mathrm{F}}$ and $\overline{y}_{\mathrm{D}}$, which arise from the absence of the lowest part of the microdosimetric spectrum due to the electromagnetic noise. Because of these characteristics and their comparably low price and easy operability, the DµDs could become the reference diamond-based systems for ion-beam microdosimetry.

Solid-State Telescope

Ion and target fragmentation are important phenomena to take into account when characterising a clinical ion beam. Their highest probability of occurrence is at the distal part of the Bragg peak for carbon ions and the use of a diamond-based two-stage telescope microdosimeter could provide a method to separate primary and secondary components. The first stage

is a thin layer of few micrometers working as a solid-state microdosimeter, similar to those studied during the previous research programme. The second stage is a thicker layer in which low-energy primary and the fragment ions release all their energy and stop. Combining the information of the two stages, the ion species could be identified particle-by-particle and the microdosimetric partial spectra for each ion species, primary and fragment, can be collected. A prototype has already been produced and preliminary measurements were performed by the groups of the National Physics Laboratory (NPL), Surrey University, and the Istituto Nazionale di Fisica Nucleare (INFN). The diamond-based two-stage telescope microdosimeter will be tested on both proton and carbon-ion clinical beams. While for protons this is measured in the lateral penumbras, for carbon ions the focus lies on the distal part of the Bragg peak.

WP 4.3 Microdosimetric Assessments for Radiobiological Studies

Microdosimeters in Cell Studies

The focus will be on the dosimetrically and microdosimetrically validated irradiation of two human chondrosarcoma cell lines (SW1353 and Cal78), within the framework of an FWF funded project (project number P32103-B). Cell survival in proton, carbon ion, and photon beams (reference irradiation) will be established in cooperation with the radiobiology group. The microdosimetric spectra will be obtained at the same water-equivalent depths where the cell samples will be placed. Multiple positions are planned for this project: plateau, mid spread-out Bragg peak (SOBP) (which is the standard position used in radiobiological studies), and fall-off. As part of the EC funded project "Heavy Ion Therapy Research Integration *plus* (HITRI*plus*)", which started in 2021, a microdosimetric investigation of the two SOBPs used for cell irradiation will be done with solid-state microdosimeters in clinical mode.

Hedgehog Filter

Irradiation of the gas microdosimeter, i.e. tissue equivalent proportional counter (TEPC), can only be performed in the so-called low-flux mode, where the particle intensity is reduced considerably in comparison to the conventional clinical mode. The low-flux mode offers no spot-scanning capability; only one single energy central spot is available. Therefore, the straightforward irradiation of a low-flux SOBP becomes impossible. As a work-around, an external filter (called hedgehog) was designed based on Monte Carlo simulations, further 3D-printed and validated. The hedgehog filter modulates single energy beams into the desired SOBP. It will be positioned in front of the phantoms during both, radiobiological cell irradiation and microdosimetric investigations. This ensures the same irradiation quality for both, the obtained relative biological effectiveness (RBE) values as well as the microdosimetric spectra. Dosimetric measurements will be performed in parallel to every cell irradiation to ensure accurate positioning of the cell flasks.

Radiobiological Modelling

Several models exist to predict a radiobiological response using microdosimetric quantities. However, the only such model that can use microdosimetric spectra as an input, and can produce a full survival curve for a variety of cell lines, is the microdosimetric kinetic model (MKM). The downside of this model is the reliance on five different radiobiological parameters,

which lack a standard operational definition. Working towards its standardisation is especially difficult as no paired radiobiological and microdosimetric measurements along a SOBP are currently available for ion-beam therapy. Dedicated measurements will be conducted towards that goal, by providing a set of inputs and reference results against which to compare possible calibrations of the MKM.

High-Precision System for Combined Microdosimetry-Radiobiology Study

In order to obtain highly accurate and reliable combined microdosimetric-radiobiological data, the highest position accuracy has to be achieved, possibly down to 50 µm. A novel method is under development; it would allow to irradiate simultaneously six cell samples at different depths and to replace each cell sample with a microdosimeter, allowing a precise matching between the cell samples positions and the microdosimetric spectra measured. Its performances will be validated in clinical environment.

WP 4.4 Microdosimetric Assessments for Clinical Applications

The acquired capabilities of representing LET distributions and mean values through experimental microdosimetry can be matched with the increasing clinical interest on LET-optimised treatment plans and, more generally, to LET-driven selection of dose.

The first step of the programme foresees dense representations of LET distributions, converted from microdosimetric spectra. Microdosimetric spectra will be collected at small longitudinal and transversal steps in a cubic target with a side of five centimetre within a water phantom. In particular, the distal part of the target will be also analysed using the microdosimetric telescope, described in WP 4.2, for the characterisation of the contributions to LET from the primary and fragment ions. This will be the foundation for the first plan of the microdosimetric commissioning of clinical beams.

In the second step, similar acquisitions will be performed in anthropomorphic phantoms and paired to the computational assessment of LET mean values as reference end-to-end test. The acquired microdosimetric spectra will also be used in an experimentally based assessment of the biological effectiveness based on the MKM.

Main Involved Personnel and Timeline

Together with the groups of the Medical University of Vienna, the TU Wien, and EBG MedAustron, which are currently collaborating, the programme on microdosimetry will be carried out in partnership with the University of Surrey and the NPL in the United Kingdom, as well as the INFN in Catania, Italy.

WP 4.1 "Uncertainty Budget for Microdosimetry Analysis" will be coordinated by Hugo Palmans (EBG MedAustron and NPL), with contributions from every institute. WP 4.2 "Advanced Microdosimetric Prototypes" will be a joint project overseen by Giulio Magrin (EBG MedAustron), with support from Hugo Palmans, Francesco Romano (INFN), and the three PhD students Sandra Barna (Medical University of Vienna), Cynthia Meouchi (TU Wien), and Gabriele Parisi (University of Surrey). WP 4.3 "Microdosimetric Assessments for Radiobiological Studies" will be coordinated by Sandra Barna in cooperation with Sylvia Kerschbaum-Gruber (Medical University of Vienna), with support from Giuseppe Schettino (University of Surrey and NPL) and the Master/PhD students Anna Falkner (University of Vienna), Katharina Stasny (University of Applied Sciences Krems and EBG MedAustron),

Jacopo Magini (University of Surrey) and Chiara de Lucia (University of Surrey). Giulio Magrin will take the role of focal person for WP 4.4 “Microdosimetric Assessments for Clinical Applications”, with support from Hugo Palmans, Sandra Barna, and Cynthia Meouchi.

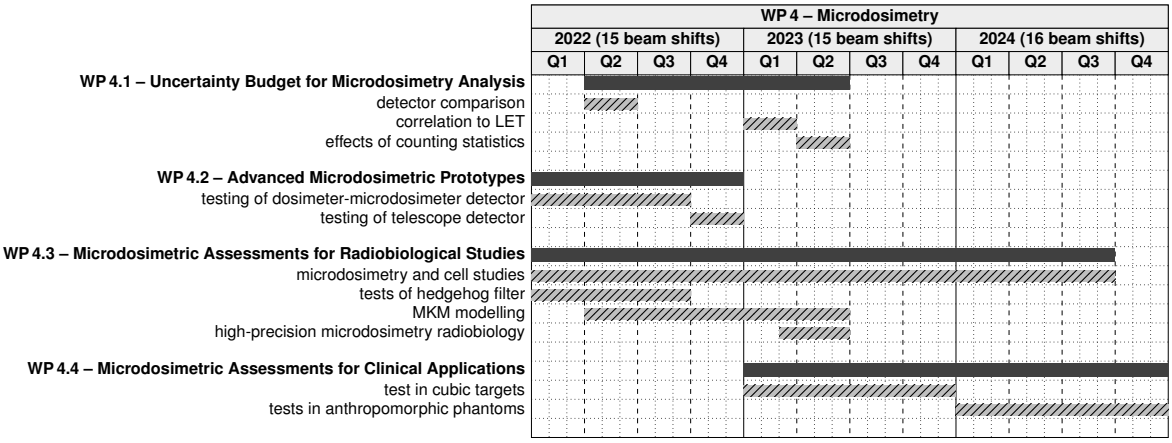

Figure 4.1: Timeline for WP 4 “Microdosimetry”.

WP 5 Laboratory Course

Objectives

- train students interested in applied particle physics or medical physics at the MedAustron research beam in a dedicated shift
- educate potential personnel for EBG MedAustron

At the TU Wien, the laboratory course *Experiments at the MedAustron Particle Accelerator – Applied Particle Physics and Medical Physics* was added to the curriculum of *Technical Physics*. Its key component will be the preparation and the implementation of a dedicated beam time at MedAustron. A small experiment, e.g. the measurement of the particle beam energy using the Peakfinder, will be prepared and the students will be responsible for data taking and subsequent analysis. Finally, the results will be presented in a lab protocol.

The laboratory course is planned as continuation of the lecture and the practical exercise *Techniques of Signal Detection and Analysis*, which was established in the curricula of *Technical Physics* and *Physical Energy and Measurement Engineering* at the TU Wien in the summer term 2020.

Main Involved Personnel and Timeline

The course will be organised and implemented by the TU Wien and the HEPHY. It will be limited to approximately eight participants due to space and logistical constraints and will take place on a yearly basis in the winter term requiring one beam shift per year. Albert Hirtl (TU Wien) and Thomas Bergauer (HEPHY) will take over the supervision of the students.

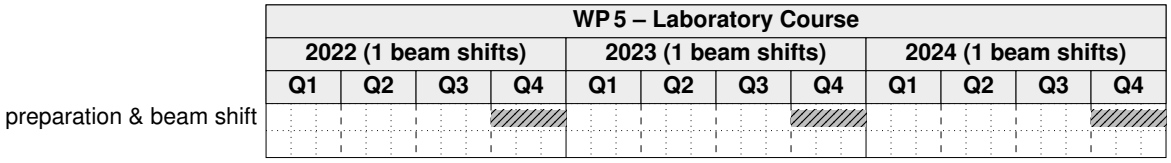

Figure 5.1: Timeline for WP 5 "Laboratory Course".

Required Beam Time

work package	2022	2023	2024	total
WP 1 – Particle Imaging	6	6	6	18
WP 2 – Instrumentation	9	9	9	27
WP 3 – Dose Determination	15	14	18	47
WP 4 – Microdosimetry	15	15	16	46
WP 5 – Laboratory Course	1	1	1	3
total	**46**	**45**	**50**	**141**

Table 5.1: Estimated beam time in shifts per year for the different work packages of "Applied Particle and Medical Physics".

Field of Research

Biophysics and Molecular Radiobiology

Coordinators:	Sylvia Kerschbaum-Gruber and Markus Zeilinger
WP coordinators:	WP 1 – Markus Zeilinger WP 2 – Sylvia Kerschbaum-Gruber/Dea Slade
Co-operators:	Medical University of Vienna, Medical University of Graz, University of Veterinary Medicine Vienna, University of Applied Sciences Wiener Neustadt, EBG MedAustron

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Research was successfully pursued within the non-clinical research unit at MedAustron in the fields of molecular radiobiology and radiobiological modelling based on in vitro experiments during the last research period. To seize unique pre-clinical research opportunities for exploiting particle radiation in cancer, the new research pillar "Biophysics and Molecular Radiobiology" was established.

Linking radiobiology research with high-quality physics and imaging data, both from a macroscopic as well as from a microscopic point of view, is a prerequisite for investigating biological effects of particle radiation. Lack of proper microdosimetry in radiobiology research and unstandardised experimental set-ups resulted in large deviations in findings reported by different groups and lack of reproducibility, which has become a growing concern in the community. To ensure robust dosimetry for radiobiology research, a strong and fruitful collaboration has been established between biology and medical radiation physics during the last period in the context of in vitro studies. For the research period 2022 – 2024, collaborative efforts will be continued and extended to include in vivo research using mouse models. Preparatory work concerning multimodality imaging of mouse irradiation has been initiated, driven by the medical physics team of the Medical University of Vienna and the University of Applied Sciences Wiener Neustadt. Our efforts shall set a global precedent for radiobiology research linked with comprehensive macroscopic and microscopic dosimetric descriptions of experimental irradiation conditions.

Radiobiology research at MedAustron will focus primarily on radiation-induced anti-tumour immunogenicity. Immune-modulatory drugs such as immune checkpoint blockade are considered the fourth pillar of cancer therapy following surgery, radiation, and chemotherapy. The principle of immunotherapy is to specifically target cancer cells by enhancing anti-tumour immune responses within the tumour microenvironment. Crosstalk between tumour cells and their microenvironment is critical for various steps during cancer progression, such as avoiding immune destruction or establishing metastasis. It is therefore critical to examine how tumour microenvironment and immune system respond to cancer therapy and how treatment strategies can be designed to induce and enhance anti-tumour immune response.

Photon-based radiation has been shown to synergise with immune checkpoint blockade in several pre-clinical models, based on stimulation of anti-tumour immunogenic signalling in cancer cells. The available data for particle irradiation is, however, scarce. This highly relevant field of research is one of the pillars of this work package. Whereas in the previous period our research focused on proton-mediated immunogenic changes in pancreatic cancer, in the upcoming period we will extend our focus to carbon ions. Carbon ions may induce different effects compared to protons due to differential ballistic properties (i.e., higher linear energy transfer (LET)). Radiation, in general, induces deoxyribonucleic acid (DNA) damage and genomic instability, resulting in the release of nuclear DNA into the cytoplasm. Cytosolic dsDNA activates the cGAS/STING pathway, which triggers interferon-dependent immune signalling and immunogenic cell death. Characterisation of the cGAS/STING-mediated immune signalling following carbon-ion radiotherapy (CIRT) with various regimens will be our major focus.

Moreover, radiation oncology can be potentiated through drugs that inhibit DNA damage response (DDR) such as poly ADP-ribose polimerase (PARP) inhibitors, as well as chemotherapeutic agents that induce DNA damage. DDR inhibitors act through the principle of synthetic lethality, whereby genetic DNA repair defects are exacerbated by drug-induced defects in a compensatory pathway. Such selective targeting of cancer cells represents a very successful example of precision medicine-driven approaches to treat cancer patients based on their

genetic background. In fact, PARP1/2 inhibitors have been approved for the treatment of ovarian, breast, pancreatic and prostate cancer with germline or somatic mutations in the homologous recombination (HR) pathway of DNA repair. In addition to killing cancer cells due to DNA damage and DNA repair failure, genotoxic stress induced by DDR inhibitors can also activate the immune system and trigger immunogenic cell death. Our plan for 2022 – 2024 is to test whether DDR inhibitors and chemotherapeutic agents can potentiate radiotherapy, particularly CIRT, by amplifying immune signalling in pancreatic cancer cells.

In addition to in vitro characterisation using cancer cell lines, in vivo mouse models will be established to verify in vitro findings. We will use a pancreatic cancer mouse model to address multiple therapy-relevant questions. We will determine the effect of radiation and combination treatments with DDR inhibitors on tumour volume and immune response.

In vivo studies will be linked with morphologic, functional and molecular imaging to enhance the visualisation of pathologies and complex biochemical phenomena. Functional and molecular imaging has become an important research field in clinical research, but is so far not utilised to its full potential in ion-beam research. In this research period, we aim to improve our understanding of the systems biology of tumours and increase the knowledge about radiobiological aspects via high-resolution multimodal imaging for tissue characterisation and response assessment.

During the last research period, the University of Applied Sciences Wiener Neustadt established the Competence Centre for Pre-Clinical Imaging and Biomedical Engineering – promoted via a large-scale research fund in 2020 – and established research infrastructure for in vivo small-animal imaging comprising micro-positron emission tomography (PET), micro-single-photon emission computed tomography (SPECT), micro-computed tomography (CT) and ultra high-field (UHF) strength micro-magnetic resonance (MR) imaging modalities in collaboration with the Medical University of Vienna at MedAustron. Research along these lines will be accelerated by the recently funded Austrian Science Fund (FWF) doc.funds.connect research grant on pre-clinical ion-beam research – "Pre-clinicAl Ion beam Research (PAIR)", a project driven by the Medical University of Vienna, the University of Applied Sciences Wiener Neustadt, and the University of Veterinary Medicine Vienna.

WP 1 Pre-Clinical Animal Research

Objectives

- methodological establishment and validation of the experimental set-up for a small-animal irradiation workflow using pre-clinical multimodal imaging techniques
- development, evaluation and verification of multi-parametric and multi-modality imaging for in vivo tissue characterisation and response assessment
- implementation of computational approaches to gather information upon the correlation between imaging biomarkers with dose and LET

WP 1.1 Small Animal Irradiation Workflow

Accurate dose prediction based on imaging is important for a small-animal irradiation process. This work package will focus on image-based treatment planning and dose calculation for X-rays and particles for small targets in mice.

To establish an magnetic resonance imaging (MRI)-only based pre-clinical workflow for treatment planning, neural network based approaches for the conversion of multi-parametric magnetic resonance imaging (mpMRI) data into synthetic computed tomography (sCT) and/or stopping powers will be explored as well as methods for automation, e.g. auto-contouring.

Treatment Planning for X-Rays

As outlined in the introduction, ongoing studies investigate optimal collimator sizes and positioning devices using the small-animal treatment planning system (TPS) for X-rays (i.e. µ-RayStation). These investigations will finish early 2022. Once the collimator set-up is fixed, basic beam data will be acquired and the beam model will be created and validated to be included in the TPS.

Treatment Planning for Particles

There is currently no dedicated TPS for particle irradiation of small animals, even though a prototype for charged particles is currently under development by RaySearch. Previous encouraging investigations of small-target irradiation at MedAustron involved the clinical TPS from RayStation v7.99 based on Monte Carlo (MC) dose calculations developed for clinical purposes. In the next step, the beam model, which was adapted for the treatment of ocular tumours at MedAustron, will be tested and validated for small-animal irradiation.

Mirco-CT

Besides the dosimetric configuration of the TPS modules, the integration of CT images and its accurate Hounsfield unit (HU) assignment needs to be implemented. In 2021, technical and physical measurements as well as computational approaches were initiated to establish an adequate methodological imaging set-up for small-animal treatment planning. Once the respective TPS environment for photons and charged particles is configured, a template plan for mice irradiation needs to be established for an efficient workflow. This comes along with a fast and automated import of the required imaging data, target delineation, and a fast and efficient collimator selection.

Dose Delivery

Based on the preliminary work, a set-up for beam modulation and mice positioning for X-ray and particle beam irradiation will be finalised in 2022. Dosimetric validation will be necessary in the design phase as well as once the system is in place The proton beam, and in the second half of this research period the carbon-ion beam, will be collimated using the passive beam modifier developed in the frame of the "Image Guided Particle Therapy of Uveal Melanoma with a Multiport Beam Arrangement (IMPACTOR)" project. The target sizes and issues tackled in this project are closely related to studies with small animals. A special frame, called *Ocularis*, was developed for eye tumour irradiation including the beam guidance and the

possibility to change collimators with different openings. This system facilitates small-target irradiation and the initial aim of this work package is to systematically test the existing frame for pre-clinical applications. Initial end-to-end testing and complete irradiation workflow with a simplified phantom are planned for the second half of 2022.

In the frame of the PAIR project, more advanced phantoms with heterogeneous structures for different tumour models will be designed and created based on 3D printing technology. Small-field dosimetry for particle beams (protons and carbon ions) and reference X-ray irradiation including MC simulations and LET and dose correlation to various imaging and molecular analysis methods will be initiated using these heterogeneous phantoms.

Corresponding studies with workflow validation and optimisation as well as end-to-end tests using the heterogeneous phantom (PAIR) and mice models are planned for 2023 in combination with WP 1.2 "Imaging-Based Dose Response". The aim is to start with radiobiological in vivo studies in 2023. Regular quality assurance (QA) measurements of the small beam line for photons and ion (including the *Ocularis* frame) beams will be necessary over the entire research period.

WP 1.2 Imaging-Based Dose Response

Besides the set-up of the dosimetric framework described in WP 1.1 "Small Animal Irradiation Workflow", an imaging framework for in vivo response assessment is the basis for all planned in vivo activities. Based on the so far successful implementation of a pre-clinical research infrastructure for in vivo small animal imaging covering micro-CT, micro-PET, micro-SPECT and UHF strength (15.2 T) micro-MR imaging modalities, the dedicated work package will cover the development and evaluation of reliable structural, functional and molecular pre-clinical imaging techniques and protocols to decipher the morphological and biological characterisation of tumour heterogeneity and normal tissue. Based on the workflow developed in WP 1.1, a MRI-only based pre-clinical workflow for treatment planning is envisaged.

The focus of the micro-MR protocol development (e.g. quantitative blood oxygenation level dependent (qBOLD), tissue oxygen level dependent (TOLD), and vascular architectural mapping (VAM)) and adaptation is on the visualisation of the dose and LET related effects of ion-beam therapy on the tumour as well as on healthy tissue microenvironment and the microvascular response. VAM will be implemented to depict the development of aberrant tumour vessels indicative of angiogenic switching. Non-invasive quantitative measurements of hypoxia using qBOLD and TOLD MR imaging sequences will be investigated. By doing so, we will be able to provide quantitative information about the tissue oxygen extraction fraction (OEF) and the oxygen metabolic rate (MRO2). Especially in the context of small volumes and sub-millimetre structures to be irradiated, geometric accuracy is of highest importance. A major prerequisite is thus to minimise MR distortion effects. Following the standards of mpMRI for clinical radiation oncology, reproducibility and standardisation of subsequent steps like image registration need to be guaranteed for the pre-clinical workflow as well. Thus, QA procedures will be established.

For all endeavours addressing the assessment of structural and morphological imaging information, specific tissue equivalent mouse and rat phantoms will be used and dedicated 3D printed phantoms will be developed. These phantoms will serve as animal surrogates and help to assure a common coordinate system and foster the improvement of imaging acquisition and reconstruction protocols. A successful image registration and spatial correlation between mpMRI and images from micro-CT will build the basis for subsequent treatment

plan creation as well as mapping with LET distributions. Image uniformity assessment will be tackled, as well as MR safety aspects. Furthermore, voxel-wise analysis of the dose and LET maps is planned for response evaluation. The required ethic applications for MRI sequence development will be submitted in early 2022, after the approval for animal housing is granted.

Depending on resources, we aim to complement MR-based imaging methods for tissue characterisation and response assessment with micro-PET imaging studies to provide an unparalleled understanding about highly specific regional, as well as systemic tumour-physiological properties with special emphasis on tumour hypoxia, metabolic tumour activity and proliferation. The specific imaging readout of the micro-PET studies will be expressed as SUV_{mean}, SUV_{max}, binding potential with non-displaceable tracer uptake (BPND) and binding potential plasma (BPP) and structurally correlated with both, the mpMRI information and with molecular tumour profiles determined in WP 2 "Translational Radiobiology" via novel machine learning based radiomics signatures.

Main Involved Personnel and Timeline

The overall work package will be coordinated by Markus Zeilinger. WP 1.1 "Small Animal Irradiation Workflow" will be coordinated by Monika Clausen and Barbara Knäusl from the Medical University of Vienna with support from Peter Kuess, Hermann Fuchs, Sabine Leitner, Markus Zeilinger, and Lukas Zimmermann. WP 1.2 "Imaging-Based Dose Response" will be coordinated by Markus Zeilinger (University of Applied Sciences Wiener Neustadt) and Barbara Knäusl (Medical University of Vienna) with the support of three PhD students, which are funded through the PAIR grant over the entire research period, as well as the scientific staff members from the Comepetence Center for Pre-clinical Imaging and Biomedical Engineering (University of Applied Sciences Wiener Neustadt) comprising three postdoctoral researchers and four scientific associates.

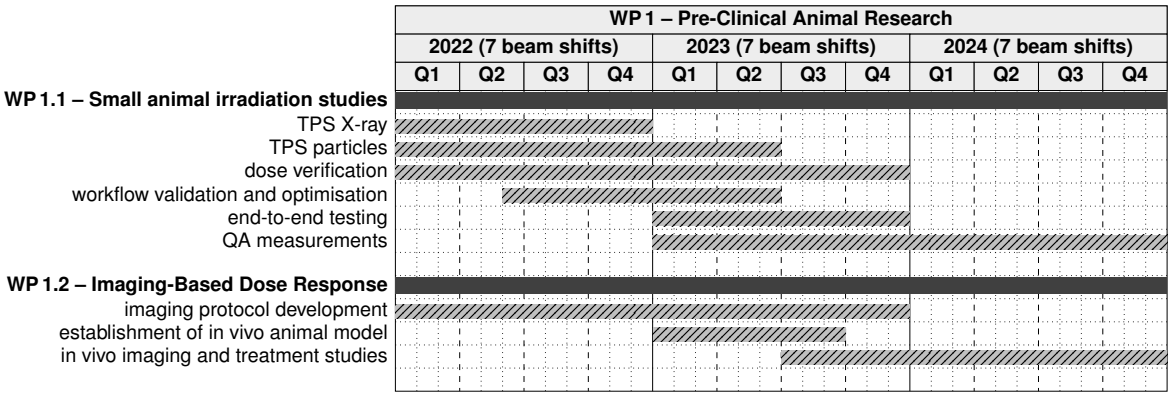

Figure 1.1: Timeline for WP 1 "Pre-Clinical Animal Research".

WP 2 Translational Radiobiology

Objectives

- elucidate key molecular players of radiation-induced anti-tumour immunogenicity and their relevance for ion-beam therapy
- characterise the synergistic potential of anti-tumour immune modulators in combination with particle therapy
- validation of in vitro endpoints in a mouse model

WP 2.1 Radiation-Induced Anti-Tumour Immunogenicity

The focus of this work package is to solidify and expand the line of research that was established in the previous research period. Namely, this work package aims at the characterisation of cancer immunogenic signalling and its potential therapeutic exploitation in response to radiation. This avenue is highly clinically relevant, as immunotherapy held great promise in revolutionising cancer therapy but failed to do so for the majority of malignancies. Radiation stimulates immunogenic signalling in cancer cells and promotes anti-tumour immunity, so far best described in breast cancer and non-small cell lung cancer. For both, pancreatic cancer and triple-negative breast cancer, we showed that a clinically relevant hypofractionation regimen of 3×8 Gy successfully instigates the Type I Interferon axis (IFN-1), which is the crucial inducer of anti-tumour immunogenicity. Single dose treatment with comparable total dose failed, while protons when investigated for the same endpoints were not superior to X-rays.

Failed IFN-1β induction in pancreatic cancer in response to single doses may be caused by dysfunctional STING phosphorylation. This work package will assess STING phosphorylation after multiple doses of carbon ions, elucidating potential LET-dependency of STING phosphorylation and activation of key mediators of the cGAS/STING pathway.

In 2022, the potential of carbon ions to induce anti-tumour immunogenicity will be investigated by focusing on key effectors of IFN-1 signalling and their potential LET-dependency. We will monitor the expression levels and/or the phosphorylation status of cGAS, STING, IRF3, TBK1, IFN-Iβ, STAT1 and several IFN-stimulated genes (ISGs) such as CXCL10, CXCL11, MX1, MX2, IFIT1 and IFIT2. Activation of ISG expression will be further analysed genome-wide by RNA-sequencing. Cancer autophagy appears to be a negative regulator of anti-tumour immunogenicity, based on the removal of radiation-induced cytosolic dsDNA. Autophagy signalling will be assessed in response to radiation with either X-rays or particles. Potential autophagy targeting strategies may be deployed in WP 2.2, based on these results.

Investigations will largely focus on pancreatic cancer. Given that BRCA-mutated cancers are genomically unstable due to HR deficiency and therefore more immunogenic, our analysis will include BRCA-proficient (PANC-1, MIA PaCa-2 and BxPC-3) and BRCA-deficient pancreatic and breast cancer cell lines (Capan-1).

Treatments that induce cGAS/STING and IFN-1 signalling were also shown to promote the expression of proteins that contribute to an immunosuppressive tumour microenvironment (TME) such as PD-L1 and galectin-1. To investigate the crosstalk between cancer cells

and TME, we will examine the expression levels of PD-L1 and the release of galectin-1 after radiation doses that are able to launch pro-immunogenic IFN-1b. Upregulation of the immunosuppressive markers would justify further efforts in targeting the immunosuppressive TME using anti-PD-L1 agents and galectin-1 inhibitors as immune checkpoint blockade. Radiation-induced immunogenic signalling coupled with immune checkpoint blockade is expected to improve treatment outcomes in pancreatic adenocarcinoma, which is otherwise refractory to immune checkpoint blockade.

WP 2.2 Combinational Strategy to Enhance Radiation-Induced Immunogenicity

Our recent research results revealed a moderate immunogenic response of cancer cells following radiation with X-rays or protons. Therefore, this work package aims at identifying combinational treatment strategies that can result in a synergistic response. Combinational approaches with DDR inhibitors have been already initiated, show promising effects, and will be further investigated in pancreatic cancer cell lines.

Among DDR inhibitors, so far only PARP1/2 inhibitors have been approved for cancer therapy including pancreatic cancer. Poly ADP-ribose polymerase inhibitors (PARPis) were shown to activate the cGAS/STING pathway to promote anti-tumour immunity in triple-negative breast cancer and small cell lung cancer. PARPi-treated cancer cells also upregulate PD-L1 expression and sensitise tumours to anti-PD-L1 immune checkpoint inhibitors, justifying the combination of PARPi with immunotherapy. In addition to PARP1/2, other PARP family members were also shown to modulate immune response, notably PARP7 and PARP14. PARP7 and PARP14 are amplified in different cancer types and may help cancer cells to overcome cytosolic nucleic acid stress by hampering immune response. Recently developed selective inhibitors of PARP7 and PARP14 were shown to induce IFN-1 signalling in melanoma and lung cancer mouse models. This opens up opportunities for new combinatorial approaches to test whether different PARPis can potentiate radiation-induced immunogenicity in pancreatic or breast cancer. Specifically, we will use the clinically relevant PARP1/2 inhibitor olaparib and the recently developed PARP7 inhibitor RBN2397 and PARP14 inhibitor RBN012759. We will extend our focus to ataxia telangiectasia mutated (ATM) inhibitors, which represent another class of DDR inhibitors that may be relevant, specifically for pancreatic cancer. The DDR kinase ATM is frequently mutated in pancreatic cancer and ATM inhibitors were shown to induce immune-relevant cGAS/STING signalling and PD-L1 expression in pancreatic cancer cells. In addition to the pancreatic and breast cancer research focus, DDR inhibition will be applied in chondrosarcoma as part of our collaborative efforts with the Medical University of Graz. Current results indicate that chondrosarcoma are largely resistant to radiation with either X-rays or protons. In 2021, research efforts already focused on carbon ions without and with additional DDR inhibitor treatment. Preliminary results indicate synergistic effects of an ataxia telangiectasia and Rad3-related protein (ATR) inhibitor with carbon ions in chondrosarcoma in vitro. This research is carried out within a running FWF grant (PIs: Birgit Lohberger, Dietmar Georg) and will continue in the period 2022 – 2024, characterising the radiation response of chondrosarcoma to particle treatment.

Beyond DDR inhibitors, this work package aims to furthermore elucidate the potential of classical chemotherapeutics and targeting of immuno-relevant cellular mechanisms such as autophagy in combination with radiation to improve anti-tumour immunogenicity. Another objective of this work package will be to combine chemotherapeutic agents with radiation to increase DNA damage and genomic instability and thereby make tumours more immuno-

genic. Specifically, we will use different chemotherapy drugs that are used in pancreatic cancer chemotherapy and that were shown to activate the cGAS/STING pathway: irinotecan (topoisomerase I inhibitor), oxaliplatin (crosslinking agent), taxol (microtubule-stabilising antimitotic agent) and 5-fluorouracil (pyrimidine analog antimetabolite).

To address the previously identified dysfunctional STING phosphorylation in pancreatic cancer cells, we will treat the cells with STING agonists to examine whether STING can be activated in combination with radiation, DDR inhibitors and chemotherapeutic agents. The synergistic effects of all aforementioned combinational treatments will be assessed as described under WP 2.1 by monitoring the induction of IFN-1 signalling at the mRNA and protein level.

WP 2.3 In Vivo Mouse Model Experiments

Radiation-Induced Anti Tumour Immunogenicity for Pancreatic Cancer

IFN-1 signalling induces immunogenic cell death via expression of cell surface markers that activate immune cells. In pancreatic cancer cells, we observed radiation-induced surface expression of calreticulin and release of HMGB1, which are well-characterised markers of immunogenic cell death. To assess the functional consequences of the expression of these markers, we will perform a cytotoxic T-lymphocyte assay. Efficient T-lymphocyte-mediated killing of cancer cells will justify further application of in vivo mouse models.

Therapeutic potential of radiation and combinatorial treatments may only be assessed using in vivo models. The MedAustron mouse facility and the multimodality in vivo imaging lab are expected to become operational in 2022, which will allow us to translate our findings from studies in pancreatic cancer cell lines (WP 2.1 and WP 2.2) to in vivo mouse models. The effect of radiation and combinational treatments on pancreatic cancer will be determined by measuring tumour volume, mouse survival, acinar to ductal metaplasia, cellular viability (proliferation, apoptosis and necrosis), tumour infiltrating lymphocytes and IFN signalling.

Within this work package, collaborative efforts with the PAIR partners (University of Applied Sciences Wiener Neustadt, University of Veterinary Medicine Vienna) will furthermore focus on cancer heterogeneity and hypoxia, which are both prevalent within pancreatic cancer and negatively impact therapy outcomes. In an effort to tackle these pertinent cancer features, we will couple mpMRI with ex vivo immunohistochemistry techniques in the pancreatic cancer mouse model established in this work package. As part of radiomics analyses, mpMRI will be performed using a dedicated 15.2 T UHF MR system. We will develop an approach that utilises blood as an endogenous contrast agent and enables non-invasive, quantitative measurement of TME hypoxia using dedicated MR imaging sequences. This approach will provide a hitherto unparalleled understanding of tumour heterogeneity and TME as a function of dose and LET, which is a visionary step for response assessment in the context of ion-beam research.

Normal Tissue Mouse Brain Model

Based on the availability of funding, we plan to investigate normal tissue response in a mouse brain model. Specifically, we will test the suitability of MRI as a non-invasive detection method for vasculopathies following brain irradiation, a prevalent normal tissue adverse reaction in glioblastoma patients. In 2023, the model will be established, including end-to-end testing. With 2023, we plan to perform brain irradiation with X-rays, protons and carbon ions with low, intermediate, and high radiation doses. Besides, in vivo and imaging-based

characterisation of dose and LET effects, we will assess morphological tissue changes, vascular oedema and DNA damage in paraffin-embedded brain tissues. Neuroinflammation markers in irradiated brains will be assessed using quantitative polymerase chain reaction (qPCR). Results will be compared between radiation modalities to determine the potential LET dependency of adverse reactions such as vasculopathies, and the MRI suitability as a non-invasive method for monitoring adverse reactions in normal tissue. An FWF grant application related to this content will be submitted in 2022.

Main Involved Personnel and Timeline

This work package will be coordinated by Sylvia Kerschbaum-Gruber and Dea Slade. WP 2.1 and WP 2.2 will be supported by two PhD students (Katharina Stasny and one additional PhD student to be recruited in 2022). A technician will be recruited in 2022 to support the establishment of WP 2.3 "In Vivo Mouse Model Experiments". Undergraduate students will continue to support biology research, as in the previous research periods, and will help to staff shifts. Furthermore, the collaboration with the Medical University of Graz will continue also with supporting the shifts. In addition, biologist from the Medical University of Vienna as well as the University of Veterinary Medicine Vienna will contribute in the future. In vivo irradiation studies are a collaborative effort between the Medical University of Vienna and the University of Applied Sciences Wiener Neustadt. The projects will be supported by all PAIR PhD students funded over the entire research period. Animal housing operation and animal welfare responsibilities are performed by all work package associated personnel with respective qualification coordinated by Dea Slade.

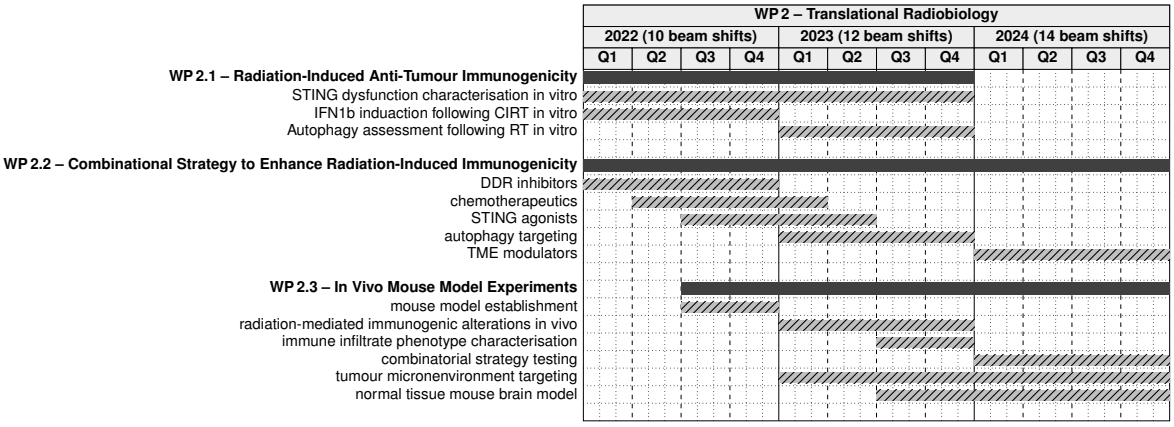

Figure 2.1: Timeline for WP 2 "Translational Radiobiology".

Required Beam Time

work package	2022	2023	2024	total
WP 1 – Pre-Clinical Animal Research	7	7	7	21
WP 2 – Translational Radiobiology	10	12	14	36
total	**17**	**19**	**21**	**57**

Table 2.1: Estimated beam time in shifts per year for "Biophysics and Molecular Radiobiology".

Field of Research

Technological Innovations and Clinical Implementation

Coordinators:	Hermann Fuchs and Barbara Knäusl
WP coordinators:	WP 1 – Barbara Knäusl WP 2 – Hermann Fuchs WP 3 – Monika Clausen
Co-operators:	Medical University of Vienna, EBG MedAustron

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This research field focuses on translational technology-oriented research aiming at clinical implementation, validation and subsequent wide scale application. Projects fostering technological innovations as well as research and development have been initiated and conducted in close collaboration with EBG MedAustron already during the previous research period, e.g. 4D dose calculation and artifical intelligence (AI)-based image processing were developed based on retrospective data.

Systematically acquired imaging data in combination with a systematically acquired (long-term) treatment outcome data is scarce in radiation oncology research. At MedAustron, large prospective clinical studies, such as the register study are currently ongoing and produce such valuable data for in-depth analysis including clinical response assessment. This is the main pillar for neural network based outcome modelling.

The aspects of retrospective analysis of clinical data are manifold, ranging from the retrospective analysis of the treatment of moving targets in a 4D dose tracking framework, to the dose accumulation of adaptive treatments and the correlation between multi-parametric magnetic resonance imaging (mpMRI) data and clinical outcome. All these aspects are highly relevant for novel adaptive radiation therapy (ART) concepts, especially when dealing with protons or carbon ions.

Based on existing treatment plans and patient specific quality assurance (QA) measurements, independent dose calculation algorithms were implemented in the previous research period. An algorithm for carbon-ion dose calculation was successfully implemented but lacks clinical commissioning.

Besides the retrospective analysis, dosimetric end-to-end tests are the basis for clinical implementation. In the context of moving targets, end-to-end tests were developed and validated in the previous research period. In this research, period these achievements are the basis for validating motion mitigation strategies of clinical interest, such as gated beam delivery or re-scanning.

Several proton and carbon-ion centres treat ocular tumours with particles since decades demonstrating good local control rates. At the Medical University of Vienna/AKH there is a long time experience with stereotactic radiation therapy (SRT) demonstrating similar local control rates as proton therapy in other countries. While almost all proton centres treat patients in seated position and with marker clips without 3D imaging, treatment planning at the Medical University of Vienna/AKH with SRT is based on 3D imaging, computed tomography (CT) and magnetic resonance (MR), and patients are treated in supine position without clips. The aim of the Austrian Research Promotion Agency (FFG) funded project "Image Guided Particle Therapy of Uveal Melanoma with a Multiport Beam Arrangement (IMPACTOR)" was to establish ocular treatments at MedAustron combining the experience from SRT at the Medical University of Vienna/AKH and the ion centres from other countries. Therefore, within this project hardware to reduce the beam size, an eye tracking device, and an eye tracking software was developed in the previous research period and will be finalised, tested, medically certified and ultimately started to be clinically used within this research period. This work will go on after the end of the IMPACTOR project, coordinated by MedAustron personnel.

Magnetic resonance imaging (MRI) already provides significant benefits for radiation therapy due to the improved soft-tissue contrast. With the development of new MR sequences it will be possible to further improve current MR-only workflows by reducing the need for parallel planning CTs. This is an extension of the MR-only workflow developments already started in the last research period. The developed methods need to be carefully evaluated in

accompanying treatment planning studies, which will also be employed to identify the benefits and limitations of a novel treatment concept, on-line MR guided particle therapy. Being able to adequately treat non-static anatomy would significantly increase the potential patient collective benefiting from particle therapy.

Finally, all aspects of maintaining as well as improving the QA programme of research equipment and the available proton and carbon-ion beams available at MedAustron are covered within this project. This encompassed on one hand a continuation of the regular QA measurements, that were acquired during the previous research period, as well as the validation of new accelerator features which will be implemented in the coming years.

WP 1 Clinical Implementation and Validation

Objectives

- framework for dose accumulation and response assessment
- translation of ocular treatments into clinics

WP 1.1 Outcome Modelling and Automatisation

Dose accumulation and its dosimetric validation is a long lasting challenge and the main source of uncertainty for clinical radiobiological modelling, which requires developments based on retrospective data in the first place. The amount of clinical data at EBG MedAustron is constantly increasing and it is planned to further utilise these data cooperation with the clinical side in studies evaluating the proof of clinical evidence of ion-beam therapy.

Clinical Dose Response Assessment

The long-term aim of these studies is the analysis of imaging bio-markers from mpMRI for dose response analysis and to implement predictive outcome parameters in the clinical treatment workflow. Radiomic data will be extracted from these images and evaluated with non-learning as well as deep-learning methods. Furthermore, anatomical information from MRI can be processed for synthetic computed tomography (sCT) generation and automated segmentation.

To optimise carbon-ion treatment planning strategies based on linear energy transfer (LET) maps, a PhD project was initialised in cooperation between EBG MedAustron and the Medical University of Vienna. The aim of this project is to improve the efficiency of carbon-ion therapy by using LET-based optimisation methods.

Independent Dose Calculation

For clinical commissioning of independent dose calculations of carbon ions, both for fixed beam lines, simulation outcomes will be compared to existing patient specific QA measurements. Long-term goal is also to replace patient specific QA measurements for patients treated with the proton gantry.

WP 1.2 Intrafractional and Interfractional Moving Targets

Retrospective Analysis

The treatment of moving targets is a long lasting challenge in ion-beam therapy. Post-processing and detailed analysis of patient data is a prerequisite prior introducing novel clinical procedures, e.g. novel particle species or reduced target margins. A 4D dose tracking framework employing the four-dimensional computed tomography (4DCT) phases, the delivery log files providing information on position and time of delivery for each spot as well as information from the breathing patterns from the surface scanner acquisition, was developed for protons in the last research period in the context of adaptive ion-beam therapy. In the upcoming research period, the automation for clinical use and expansion towards carbon ions is planned. Such a framework is not only crucial for retrospective in silico analysis of the dose distribution in a moving anatomy but also for prospective dose prediction and for end-to-end test procedures. The necessity and applicability in a clinical environment will be investigated for a variety of selected indications. Finally, the analysis of clinical data will support the end-to-end test developments for adaptive radiotherapy for interfractionally moving targets. The recently approved European Commission (EC) funded project "Real Time Adaptive Particle Therapy of Cancer (RAPTOR)" (cf. WP 3.1 "Time Resolved Dosimetry") aims for respective developments that will support future multicentric particle therapy studies. The data to be analysed within the RAPTOR project range from various imaging data, namely anatomical and functional CT and MRI data as well as X-ray based verification images, to treatment plans and treatment records.

Implementation of Novel Workflows

Expanding the clinical treatment sites towards moving tumours requires technical developments which need to be validated and continuously integrated. Besides intrafractionally moving targets also changes in between the fractions require technical innovations in order to improve the treatment accuracy as well as the clinical workflow. The pulsed beam delivery from the synchrotron accelerator poses additional challenges for the implementation of commonly known mitigation techniques, like gating or (active) breath hold. The outcome of the last research period provides all required tools for validation and integration of such novel techniques. Once gated beam delivery is enabled in irradiation room 1 (IR 1), workflow tests for gated beam delivery will be performed followed by the translation into irradiation room 2 (IR 2). Clinical decisions on the treatment technique for the selected moving indications will define final workflows. This can range from free breathing over (ventilated) deep-inspiration breath-hold to gated beam delivery.

In this research period, the design of novel clinical workflows for intra- and interfractionally moving targets will be supported, and end-to-end tests performed employing the dosimetric tools developed and optimised in WP 3.1 "Time Resolved Dosimetry". In the end, the fully established end-to-end test procedure for intrafractionally moving targets will be performed clinically at MedAustron and at all other facilities that are included in the EC project RAPTOR.

Motion assessment is a key factor to implement highly accurate motion mitigation. While surface motion assessment is fast and does not deliver any additional dose, real-time X-ray imaging during treatment provides the ground truth of the tumour motion. The Medical University of Vienna will continue to solve the problem of the correlation between the internal

and external breathing signal providing long-standing expertise in 2D/3D image registration.

WP 1.3 Technology for Ocular Treatments

The treatment for ocular tumours will go into clinical operation in this research period. The IMPACTOR project will end with March 2022 with the final report for the FFG. The research conducted for the clinical implementation of eye treatments at MedAustron will be continued in close cooperation between the Medical University of Vienna and EBG MedAustron. In the beginning of 2022, the fine-tuning of the treatment planning system (TPS) for the dedicated eye beam line in IR 1 will be finalised. In addition, gating functionality will be finally tested in IR 1 in the beginning of 2022 in combination with the in-house developed eye tracking software that is part of the IMPACTOR project. The medical product MedAustron treatment environment for ocular tumours (MATEO) including the passive beam modifier, the gating software and the eye tracking device will be freezed in the first quarter of 2022. The technology developed and validated in IR 1 will then be translated to the clinical IR 2 in the beginning of 2022. This includes assistance for developing and optimising dedicated QA and patient specific QA. Simultaneously, a cooperation between EBG MedAustron and the Medical University of Vienna will setup and establish a treatment workflow. Once the proton implementation will be up and running, the eye treatment system might be extended for other beam directions or heavier ions than protons and will be available as tool for other research projects at MedAustron.

In contrast to other centres, patients will be treated in supine position and without clips, as it is done at the Medical University of Vienna/AKH. The long term goal is to use 3D image based treatment planning without clips. Therefore, further research will focus on positioning and target delineation accuracy in a clipless workflow. Initially, the patient cohort eligible for treatment at MedAustron will be the cohort that is eligible to SRT at the Medical University of Vienna. A plan comparison study of ocular treatments between SRT and proton treatment is foreseen for the initial patients. The potential increase of the eligible patient cohort, i.e. lowering the tumour size and location requirements, will be investigated starting end of 2023.

Main Involved Personnel and Timeline

WP 1.1 “Outcome Modelling and Automatisation” and WP 1.2 “Intrafractional and Interfractional Moving Targets” will be coordinated by Barbara Knäusl (Medical University of Vienna). WP 1.1 will be supported by Peter Kuess (Medical University of Vienna) and Monika Clausen (Medical University of Vienna). Furthermore, three PhD students, namely Franciska Lebbink funded until the end of 2022, Jacob Brunner and Mansure Schafasand funded for the whole research period. WP 1.3 “Technology for Ocular Treatments” will be driven and coordinated by Andreas Resch for the first months of this research period and then continued by Patrick Roisl (EBG MedAustron) after IMPACTOR ends. Moreover, additional funding for the clinical implementation and improvement is currently negotiated.

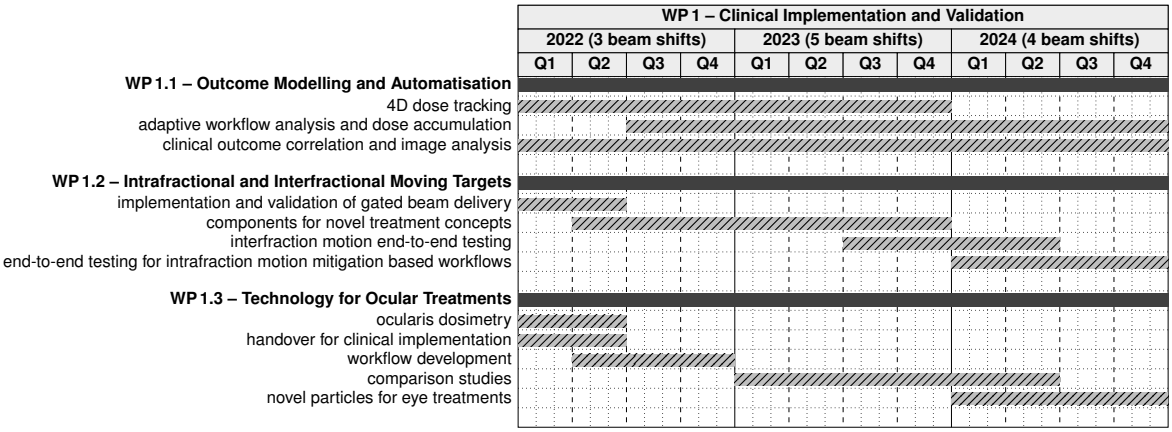

Figure 1.1: Timeline for WP 1 "Clinical Implementation and Validation".

WP 2 MRI in Particle Therapy

Objectives

- evaluation of treatment planning options for MR-guided particle therapy

WP 2.1 New Developments for MR-Only Workflow Development

A key component for using MR-only workflows is dosimetric and geometric accuracy. Established dose calculation methods in particle beam therapy rely on accurate material properties and density maps for stopping power calculations. An approach allowing to use existing systems and knowledge is to convert MRI data into sCTs, which in turn are processed using conventional implementations. So far, most sCT generation approaches are based on neural networks. In the framework of the last research period, we developed and validated such a neuronal network. A drawback of such methods is that they require realistic training data as input for training and that generation of such data sets is resource intensive. Changes in imaging routine or patient anatomy typically require additional training data for best performance. Particle therapy is a highly individualised therapy, where significant anatomy changes between patients are possible, such as e.g. removal of tissues or the insertion of stabling material. Consequently, the achievable precision is limited. In this research period, we will investigate whether additional MR techniques allow to change from mostly structural based conversion methods to methods also relying on quantitative information. Such techniques could potentially also provide additional material information, help to provide more consistent conversion results, as well as reduce the amount of training data required. In addition, these techniques may help to enhance the usefulness of MRI for clinical applications by providing additional tumour information. Recent trial results indicate that such additional data enables more personalised treatments, consequently improving treatment efficacy. A highly promising multi-parametric and quantitative MRI technique is magnetic resonance fingerprinting (MRF), which will be another focus in this work package. Only limited beam time

will be required to experimentally determine dosimetric differences of the various approaches to generate sCTs from MR data sets. The generation of sCT images is a research field that still lacks findings in the context of particle therapy, especially for carbon-ion therapy.

WP 2.2 Treatment Planning in Magnetic Fields

In contrast to photon therapy, the therapeutic beam applied in magnetic resonance guided particle therapy (MRgPT) directly interact with the magnetic fields from the MR system. Consequently, the presence of magnetic fields and their non-negligible impact on beam trajectories will require adaptions in treatment planning. So far, only few proof-of concept treatment planning studies have been performed. A clear limitation up to now was the availability of suitable treatment planning systems for MRgPT. We established a research cooperation with RaySearch (Stockholm, Sweden), the vendor of the treatment planning system RayStation, focusing among others on the development of the first commercial magnetic field capable treatment planning system RayStation 9A (RaySearch, Stockholm, Sweden).

In a first step, the implementation of magnetic field dose calculation and optimisation of an advanced TPS will be validated. A detailed benchmarking of the TPS against independent dose calculations using Geant4 application for tomographic emission (GATE)/Geant4 and experimental measurements will be conducted. After commissioning, performance studies will be executed. In a second stage, a treatment planning study is foreseen on a group of representative patient cases with malignancies located in anatomical regions where on-line MR-guidance could offer a potential benefit. The study will be conducted following the clinical recommendations from MedAustron and using data of patients undergoing treatment at MedAustron. Different target sizes and depths will be used to analyse the performance of the proposed workflow on a wider range of scenarios. The study will aim to investigate the treatment plan complexity, planning times and to identify potential benefits and/or pitfalls of the use of bended beams, for example, while sparing organs at risk (OARs). A cooperation with the Centre for Radiation Research in Oncology (OncoRay) in Germany was initiated in 2021 aiming, among others, to establish a workflow and QA process for MR guided proton therapy.

Main Involved Personnel and Timeline

Hermann Fuchs (Medical University of Vienna) will coordinate all activities assigned to WP 2 “MRI in Particle Therapy”.

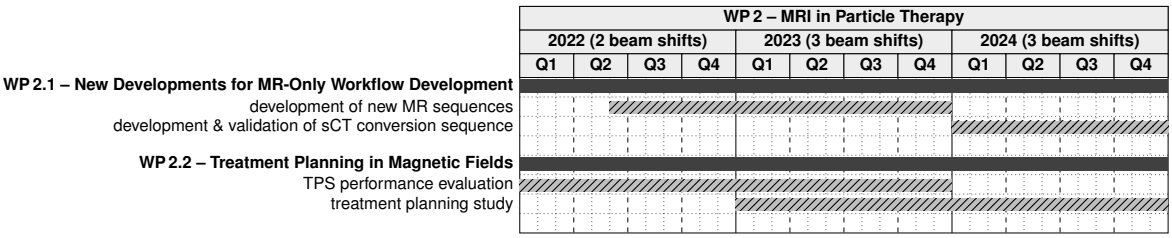

Figure 2.1: Timeline for WP 2 “MRI in Particle Therapy”.

WP 3 Quality Assurance

Objectives

- continuous assessment of beam parameters and regular equipment QA

Regular QA procedures are a prerequisite for reliable research and development activities and need to be continued throughout the whole next research period. The QA procedures introduced in the last research period will be performed in the IR 1, in regular intervals to ensure that the characteristics of protons and carbon-ion beams in the clinical beam configuration are within the established tolerance limits. Absolute dose determination and assessment of position and size of the beam will be performed in more frequent intervals, whereas beam ranges of various energies are acquired less frequently, i.e. typically once per year. In case of updates on the accelerator side, QA of the beam parameters might need to be measured earlier than planned. For such cases, a short QA programme was developed, which can be performed within another shift. In general, QA procedures will be continuously improved and adapted. Furthermore, equipment QA, either regular checks, e.g. detector cross-calibration or tests after repairs, require beam time. Five shifts are required in 2022 and four shifts per year for the reaming research period.

Main Involved Personnel and Timeline

Monika Clausen will coordinate the QA, but the measurements will be performed by all involved PhD and postdoc researchers of the Medical University of Vienna, the Technische Universität (TU) Wien, and the University of Applied Sciences Wiener Neustadt.

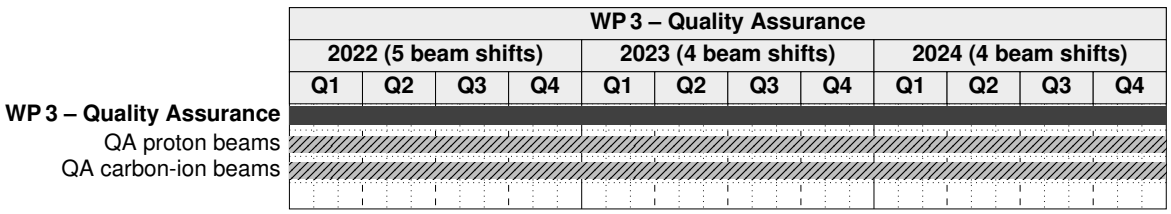

Figure 3.1: Timeline for WP 3 "Quality Assurance".

Required Beam Time

work package	2022	2023	2024	total
WP 1 – Clinical Implementation and Validation	3	5	4	12
WP 2 – MRI in Particle Therapy	2	3	3	8
WP 3 – Quality Assurance	5	4	4	13
total	**10**	**12**	**11**	**33**

Table 3.1: Estimated beam time in shifts per year for "Technological Innovations and Clinical Implementation".

Field of Research

Accelerator Physics

Coordinators:	Michael Benedikt and Claus Schmitzer
WP coordinators:	WP 1 – Nadia Gambino WP 2 – Dale Prokopovich WP 3 – Claus Schmitzer
Co-operators:	Technische Universität (TU) Wien, CERN, EBG MedAustron

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Required Beam Time . . . 76

The research field on "Accelerator Physics" was launched in the last period after the leading professor was appointed at the TU Wien. The two work packages on "Implementation of Helium Ions" and "Extraction Mechanisms" can be regarded as a continuation of the last years and essential progress is expected until the end of 2024. The third work package on "Novel Developments" provides the opportunity to deal with current topics in ion-beam therapy and in particular with the implication in accelerator physics.

The commissioning of helium ions at MedAustron is a prerequisite to explore the potential of this new particle species for both, non-clinical and clinical applications. Due to their increased mass with respect to protons, the lateral spread of helium ions caused by multiple Coulomb scattering is by a factor two lower than for protons of the same range. In addition, heavier ions exhibit greater biological damage and consequently, the biophysical properties of helium ions are intermediate of the nowadays clinically administered protons and carbon ions. Thus, the application of helium-ion beams provides a distinct clinical advantage, i.e. favourable dose distributions with attributes such as a sharper Bragg peak and lateral penumbra compared to protons, and similar potential for tumour control with a substantially reduced fragmentation tail compared to carbon ions. These unique characteristics fostered the interest of the non-clinical users at MedAustron a long time ago, which is well documented in the White Book on "Research Opportunities for Medical Radiation Physics and Radiation Biology", which was published already in 2010, as well as in the previous two documents describing the "MedAustron Non-Clinical Research Strategy" for the years 2016 – 2018 and 2019 – 2021. In a European context, helium ions are already provided for research purposes since several years at the Heidelberg Ion-Beam Therapy Centre (HIT) in Germany and a clinical programme using therapeutic helium-ion beams started recently with the first patient irradiated in July 2021. In 2020, preparation work for enabling the acceleration of helium ions was also launched at the National Centre of Oncological Hadrontherapy, i.e. Centro Nazionale di Adroterapia Oncologica (CNAO), in Italy.

Third-order slow-resonant extraction is a highly complex but nevertheless widely used technique in many accelerator centres. There are many different techniques of exciting the beam and setting up the extraction resonance, each of which leads to specific dynamic properties in the six-dimensional phase space of the extracted beam, sometimes time varying over the extraction process. For medical machines, in particular, the full understanding and control of the extraction process is a prime ingredient for successful and safe operation. The planned activities will continue on with the Conseil Européen pour la Recherche Nucléaire (CERN) collaboration to determining the optimal slow extraction method for the production of ion beams, which can be adapted for future projects such as multi-energy extraction, improvement of carbon-ion performance, and other ion beams such as helium. The results of the work package on "Extraction Mechanisms" will be of great interest to all accelerator centres employing slow-resonant extraction and relevant in general for the field of accelerator physics.

The synchrotron-based accelerator at MedAustron was designed to ensure sufficient flexibility in order to investigate the implementation of future novel developments in ion-beam therapy. In the course of time, the requirements for both, clinical and non-clinical users may change and in addition, a further development of accelerator components can be observed. Taking into account current proposals, which are discussed at dedicated international symposia and conferences such as the "FLASH Radiotherapy and Particle Therapy Conference", several topics will be investigated with an academic approach first considering the available technical infrastructure at MedAustron.

WP 1 Implementation of Helium Ions

Due to the high interest to use helium ions for non-clinical research applications, this new particle species will be primarily commissioned for delivery in irradiation room 1 (IR 1) at MedAustron. In order to explore the full theoretical properties in comparison to the other two ion species currently used, the same beam parameters will be provided. First of all, similar penetration depths of the particles in water should be available as compared to protons. This translates also to similar kinetic energy values, since helium ions exhibit a comparable relation between range and energy like protons. Figure 1.1 shows the range in water as a function of the kinetic energy per nucleon for protons and helium ions.

With the definition of the required energy range, the consequential magnetic beam rigidity is determined, which is a main parameter in accelerator physics. Other than the range-energy relation, the values of the magnetic beam rigidity are higher for helium ions in comparison to protons for the same energy values and comparable to those values of carbon ions. Figure 1.2 shows the correlation between the magnetic beam rigidity and the kinetic energy per nucleon for protons and helium ions in an energy region up to 250 MeV per nucleon. Since the synchrotron of MedAustron can accelerate carbon ions to a maximum of 400 MeV per nucleon, which corresponds to a maximum magnetic beam rigidity of about 6.4 T m, there is the potential to increase the values of helium ions up to the same values of carbon ions, i.e. a maximum kinetic energy of 400 MeV per nucleon for helium ions, which corresponds to a penetration depth in water of about 816 mm. These increased values for the energy and the penetration depth would be preferably, especially for using helium ions in the particle imaging project. A comparison of the ranges in water as a function of the magnetic beam rigidity is shown in Figure 1.3 for protons, helium ions, and carbon ions.

The total number of particles for a fully extracted spill shall be independent of the extracted energy and shall be at least 8×10^8. Even higher particle numbers, i.e. above 10^9, are preferable. For non-clinical research applications, a reduced particle flux in the range of 10^3 to 10^6 particles per second is required. This requirement is similar to the low flux settings for protons, which are already available and operational for non-clinical research in IR 1.

Objectives

- availability of a fully commissioned helium-ion beam in the synchrotron for final acceptance in IR 1

WP 1.1 Injector Commissioning

The injector commissioning includes the source branch, the low-energy beam transport (LEBT) and the linear accelerator (LINAC). First activities for the commissioning of the third ion source (S3) branch in the injector hall, which is now dedicated to helium ion operation, and the further transport up to the linear accelerator were already carried out in 2021. The S3 and the first sector of the LEBT (LE-00) is fully decoupled from the MedAustron particle therapy accelerator (MAPTA) operation, i.e. from the clinical machine. Therefore, the full source branch, all magnets of the LE-00 sector and all beam diagnostics devices (excluded only one wire scanner), were transferred in the so-called “test system”. The test system is not connected to MAPTA and therefore we had the advantage to perform some parallel activities

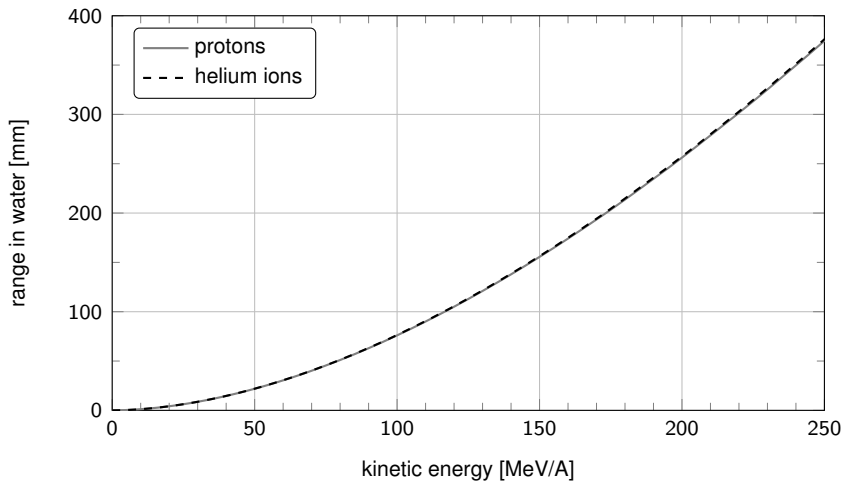

Figure 1.1: Ranges as a function of the kinetic energy for protons and helium ions.

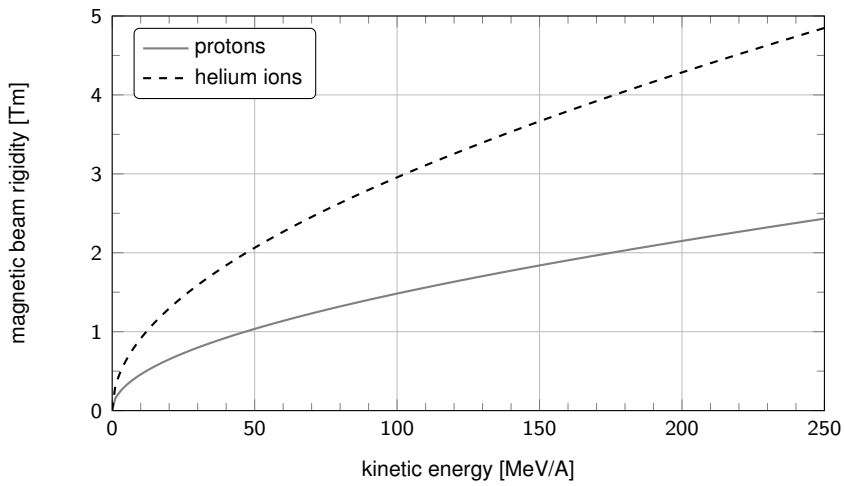

Figure 1.2: Magnetic beam rigidity as a function of the kinetic energy for protons and helium ions.

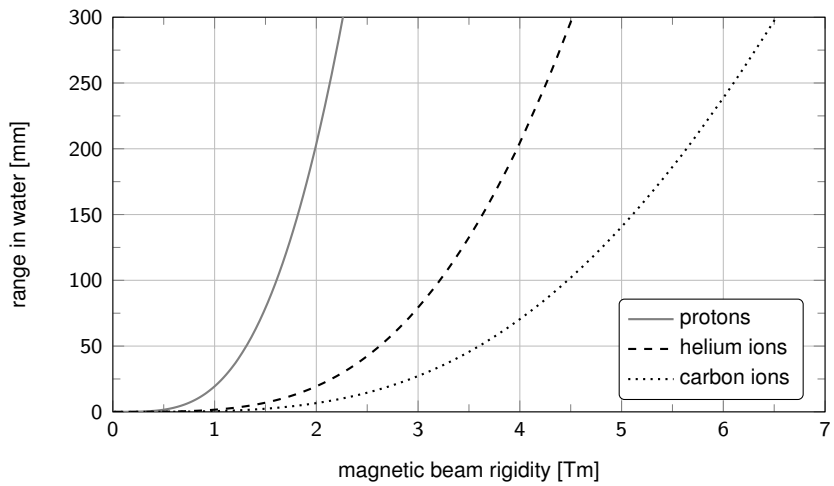

Figure 1.3: Ranges as a function of the magnetic beam rigidity for protons, helium and carbon ions.

to clinical treatment in the S3 branch. The main milestone reached was to put into full operation the source with all its subcomponents, the beam diagnostics devices in the source and LE-00 section, and all magnets. Moreover, the commissioning of a stable helium ion set-point provided by the source was also performed with the characterisation of the beam in terms of intensity, stability, and emittance at the source level. We also optimised the beam transport from the source to the LEBT with a transmission of almost 90 %. We moreover managed to transport the beam up to the radio-frequency quadrupole (RFQ) with some preliminary transmission measurements from the LEBT to the RFQ. Currently, beam optimisations in the LEBT section are taking place in order to optimise the beam steering and transport to the LINAC. The challenging part of the injector commissioning is the adaptation of the charge to mass ratio of helium ions ($q/m = 1/2$) in comparison to the one of protons or carbon ions ($q/m = 1/3$).

WP 1.2 MEBT Commissioning

The medium-energy beam transport (MEBT) commissioning is foreseen to start in 2022 and the commissioning strategy will be very similar to the other beam lines. In total, 12 shifts are required for the MEBT commissioning.

WP 1.3 Main Ring Commissioning

The main ring commissioning is foreseen to start in 2023 depending on the priorities with the other projects and the advancement that can be reached with the allocated helium-ion shifts. For the main ring commissioning, we foresee a similar strategy as applied for the other beam lines and a similar amount of shifts as required for the carbon-ion commissioning. The main commissioning tasks are:

- injection into the synchrotron;
- first turn injection;
- first acceleration;
- capture/acceleration optimisation;
- tune/emittance measurements;
- extraction.

It is foreseen to request about 42 shifts. Some of the main beam diagnostics devices will also need to be adapted in term of settings (gain, exposure time, etc.) in order to measure a proper helium-ion beam.

WP 1.4 HEBT Commissioning and IR 1 Acceptance

The high-energy beam transport (HEBT) and IR 1 commissioning is foreseen for 2024. The strategy for the HEBT commissioning will be similar to the one applied for the other particle species. The main tasks for the HEBT commissioning are the following:

- clinical energies to the IR 1 beam line;

- extraction optimisation;
- dispersion closure;
- HEBT steering;
- horizontal bending (dipol) magnet (MBH) achromatic bend;
- spot size adjustments in IR 1;
- fine steering of the beam.

For the HEBT and IR 1 commissioning, 25 shifts are intended to be used. By 2025, the helium-ion beam is foreseen to be commissioned in IR 1 and the following acceptance tests will finalise the commissioning project for helium ions in IR 1.

Main Involved Personnel and Timeline

Various experts of the accelerator team at MedAustron will be involved in the commissioning activities and will act as supervisors for diploma and project students. In addition, a postdoc position at the TU Wien is intended to be filled in the beginning of 2022 in order to support all planned research activities in accelerator physics.

Figure 1.4 summarises the different phases of the commissioning activities, which are required to provide helium-ion beams for non-clinical research applications in IR 1. The chart shows the duration of the single tasks and the completion of the corresponding activities is indicated as a milestone. The total beam time requested for the commissioning of helium ions in the years from 2022 until 2024 sums up to 76 shifts at eight hours. According to the displayed timeline, the final acceptance tests for helium ions in IR 1 will be performed in the beginning of the next research period.

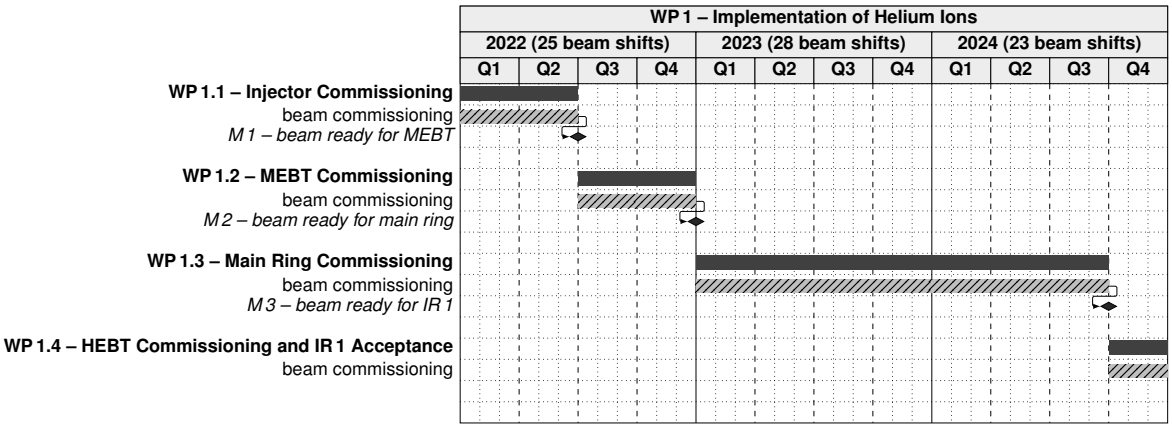

Figure 1.4: Timeline for the work package on "Implementation of Helium Ions". First activities were already launched in 2021. Due to the considerable beam time needed for providing a fully commissioned new particle species at the synchrotron-based facility at MedAustron, the final acceptance measurements in IR 1 will be carried out in the beginning of the next research period, i.e. during the year 2025.

WP 2 Extraction Mechanisms

Third-order slow-resonant extraction is a highly complex but nevertheless widely used technique in many accelerator centres. There are many different techniques of exciting the beam and setting up the extraction resonance, each of which leads to specific dynamic properties in the six-dimensional phase space of the extracted beam, sometimes producing time variations over the extraction process. For medical machines, in particular, the full understanding and control of the extraction process is a prime ingredient for successful and safe operation.

Objectives

- understanding the mechanisms behind optimisation of the slow extraction mechanisms being investigated and improving the beam quality to the irradiation rooms

WP 2.1 Slow Extraction Optimisation

There are alternative methods of third-order resonance extraction techniques in addition to the Betatron core extraction technique used at MedAustron. Development of these novel extraction techniques alongside comprehensive understanding and optimisation of the existing extraction methods can lead to improvements in accelerator operation.

This programme is aligned to a research agreement with CERN covering, on a more general basis, research on slow-resonant extraction techniques including developments of novel slow extraction methods such as constant optics slow extraction (COSE). Novel extraction methods such as COSE, which involves adjusting the lattice optics to move the resonance towards the beam stack with fixed "beams-eye" optics or Q-Sweep, which moved the tune during extraction to drive the extraction process (as opposed to moving the beam momentum towards the beam stack in Betatron driven extraction) will be investigated. Radio frequency (RF) noise driven extraction will also be investigated where the beam is spread and driven in the momentum space towards the resonance by the synchrotron RF cavity. The alternative slow extraction techniques have some advantages over the present extraction method where they can be potentially more flexibly applied to the extraction process with potential for better beam quality, fast controlled extraction, and multi-energy extraction (MEE).

In order to optimise the beam for any slow extraction method, a thorough investigation into fundamentals of the systems performance is required, including understanding of how power converter current ripples in the different magnet families affect the extracted beam quality. Investigation into the transfer mechanisms and mitigations for intensity and position ripples have been studied for Betatron core extraction and the further investigation of the beam quality for different extraction methods will be the main focus.

The work package will incorporate detailed simulations of the slow extraction methods and as such the simulations will be validated and optimised using measurement data. As part of the simulation to be performed, different simulation toolkits (MAD-X, WinAgile, MapTrack, Monte Carlo methods, etc.) are to be used in order to take advantage of, and validate, the strengths of each package where possible.

The results of these studies will be of great interest to all accelerator centres employing slow-resonant extraction and relevant in general for the field of accelerator physics.

WP 2.2 RF Knock-Out and Multi-Energy Extraction

The beam is presently extracted from the MedAustron synchrotron using the third order slow-resonant extraction technique inductively momentum driven by a Betatron core magnet towards the resonance. This method to extract the beam requires an unbunched coasting beam with a large momentum spread, which is not suitable for the re-bunching and re-acceleration required to perform MEE. Presently, if beam is still present in the synchrotron and another energy is clinically required, in order to generate another energy the remaining beam is dumped and a whole new acceleration cycle is required.

Radio frequency knock-out (RFKO) is an alternative extraction method that uses a transverse RF signal to excite the beam in amplitude while still bunched for slow extraction. The advantage of using RFKO is that the extraction can be stopped quickly by cutting the RF signal to the transverse exciter and adjusting the beam energy for the remaining circulating beam by adjusting the synchrotron RF frequency before recommencing extraction. Another advantage of RFKO is that with continued feedback from the beam diagnostics, the intensity of the extracted beam can be controlled for dynamic intensity control so that high and low intensity weighted spots can be requested to produce a more controllable dose delivery within the tens of microsecond regime. Presently, with the slow response of the Betatron core extraction, the intensity can only be varied in the order of tens to hundreds of milliseconds.

This work package aims to simulate and test the feasibility of implementing RFKO extraction at MedAustron for improved beam quality, MEE, and dynamic intensity control. In parallel to theoretical simulation analysis, the results will be continuously benchmarked with machine measurements using existing systems and, where possible, flexible prototyping hardware such as a software defined radio (SDR).

RFKO requires detailed knowledge of beam dynamics with particle tracking using dynamic lattice elements alongside the evaluation of the effects generated by transverse RF on the emittance and longitudinal dynamics of the beam. The simulated series of configurations will be evaluated based on the spill quality, extraction efficiency, spot shape, spot size, momentum spread, and intra-spill extraction beam parameters. A new set of machine operation parameters will be required for RFKO and these will be simulated for optimal, on and off momentum, extraction efficiency and validated with tune and chromaticity measurements along with momentum spread measurements to validate the required configuration before experimental validation.

WP 2.3 Phase Space Tomography

Phase space tomography involves the measurement of extracted beam though a rotation in the phase space in the HEBT beam line after the two elements designed to rotate the bar of charge, the phase space stepper (PSS), and the gantry rotator. This work is a continuation of the PhD thesis completed in February 2020 by Alexander Wastl.

Updated measurements will be collected using the reduced emittance from the synchrotron after PIP1 b and will include measurements with both the PSS and the gantry rotator for fine phase space rotations with different reconstruction methodologies and comparison to simulated phase space tomography responses. Initial reconstruction of the bar of charge will be performed with protons however once the optimal tomographic reconstruction is determined then carbon ions and potentially helium ions could be investigated.

WP 2.4 PIP 1b

The performance increase activities for providing extended accelerator functionalities for beam delivery in IR 1 started already in 2021. The current upgrade includes the optimisation of the slow extraction mechanism with RF channelling to reduce ripples at higher intensities and reduce the emittance in the ring to optimise the extracted beam parameters. With the implementation of these new settings for IR 1, all beam lines at MedAustron can be operated using the same PIP 1b configuration. Additional features for the non-clinical users will be considered in order to further decrease the irradiation time for some special applications, i.e. reduced extraction times in comparison to the nominal 10 s.

Main Involved Personnel and Timeline

An overview of the intended activities is summarised in Figure 2.1 for all three sub-work packages. Only WP 2.4 "PIP 1b" will be finalised during the upcoming year 2022. The three remaining sub-work packages will last until the end of this research period.

The work package on "Extraction Mechanisms" involves an ongoing collaboration between the TU Wien and MedAustron as well as a collaboration agreement between CERN and MedAustron signed in 2020. The present PhD student Florian Kühteubl (TU Wien) will continue his PhD project started in October 2020 on slow extraction optimisation (inclusive WP 2.1 and WP 2.2). Cividec will also provide support for measurements related to ripple studies (Erich Griesmayer is a co-supervisor of Florian Kühteubl).

Main involved personnel are (in alphabetical order):

- Pablo Arrutia Sota (CERN)
- Matthew Fraser (CERN)
- Erich Griesmayer (Cividec and TU Wien)
- Florian Kühteubl (TU Wien)
- Mauro Pivi (MedAustron)
- Dale Prokopovich (MedAustron)
- Alexander Wastl (MedAustron)

WP 3 Novel Developments

Objectives

- exploration of accelerator impacts for novel approaches in ion-beam therapy

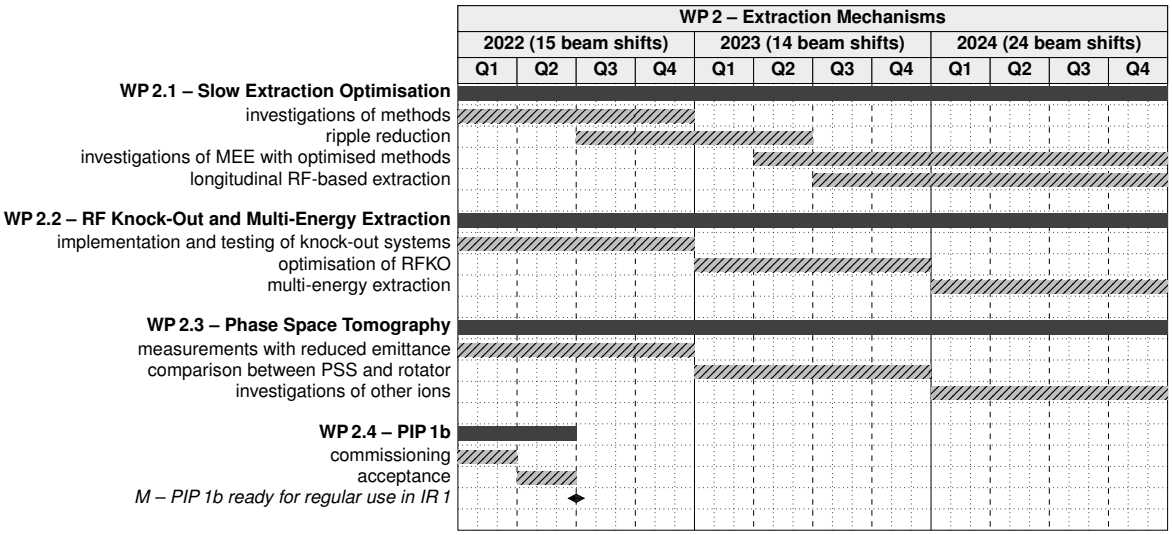

Figure 2.1: Timeline for the work package on "Extraction Mechanisms".

WP 3.1 Ultra-High Dose Rate (FLASH) Radiotherapy

Treatments with high dose rates, i.e. larger than 40 Gy/s, have repeatedly proven to spare healthy tissue during treatments while preserving the destructive antitumour effect. Ultra-high dose rate (FLASH) radiotherapy could be executed faster and thus with a lower probability of patient movement in comparison to standard treatments with a dose rate of 2 Gy/s. So far, there is no clinical proton or carbon-ion centre, which offers standard FLASH treatments on human patients, as the underlying effects still lack founded explanations. MedAustron would be capable of FLASH compatible dose rates, if the Betatron extraction is executed at timescales around 10 ms to 50 ms. This is expected to affect the time structure but not the extracted energy or transverse beam properties, thus will not require HEBT recommissioning. In a first step, an appropriate set-up of the synchrotron for such high dose rates has to be prepared and the time structure will be verified on the quality intensity monitor. As a next step, the IR 1 isocentre shall be irradiated and a passive dosimetry device, such as radiochromic films or thermoluminescent dosimeters (TLDs), can measure the total applied dose while a reconfigured dose delivery system monitors the start and stop times of the extracted spill. In a theoretical description, a typical FLASH treatment set-up can be described and typical doses of FLASH fractions are estimated, especially the number of spills required for medically meaningful doses. The study shall comprise the possible application using a three-dimensional range shifter and a proposal of a widened beam optics to irradiate the full tumour via a passively scattered beam.

WP 3.2 Mixed Beams

Typical particle treatment facilities monitor the transverse beam properties and the applied dose during treatments but tend to neglect the online monitoring of range and changes in tissue density due to its complexity. Instead, treatment plan quality assurance (QA) is executed regularly using water phantoms and QA detectors as well as periodic checks of patient computed tomography (CT) images. The application of a mixed particle beams consisting

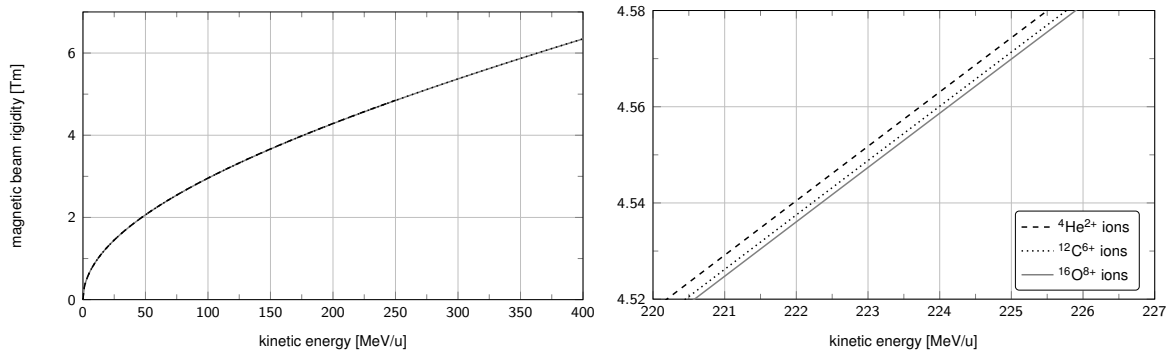

Figure 3.1: Magnetic beam rigidity as a function of the kinetic energy per unit mass for helium ions ($^4He^{2+}$), carbon ions ($^{12}C^{6+}$), and oxygen ions ($^{16}O^{8+}$) exhibiting a similar but not equal charge-to-mass ratio of $q/m = 1/2$. The differences in the values for the magnetic beam rigidity can be seen in the zoom in for the energy range of 220 MeV/u to 227 MeV/u in the right diagram.

of very light and heavy particles could provide a solution to treat patients with carbon-ion beams while exploiting the longer range of helium ions for online diagnostics in a dedicated detector set-up. Thus, the carbon-ion beam must contain a trace amount of helium ions of the same charge to mass ratio. The currently deployed electron cyclotron resonance (ECR) ion sources at MedAustron are theoretically capable to provide traces of $^{12}C^{6+}$ and $^4He^{2+}$ beams, which are co-extracted simultaneously and could be used for investigations towards such an application. Unfortunately, the $^{12}C^{6+}$ production rate in a 14 GHz ECR ion source is very low (some µA) and would be overshadowed by $^4He^{2+}$ production (about 0.5 mA). Yet the current set-up would allow a mixed $^{12}C^{4+}/^3He^{1+}$ beam, when using ^3He gas instead of the currently used ^4He gas.

In this configuration, preliminary beam separation tests and new mixed beam emittance measurements analysis methods can be tested in the existing S3 branch. Furthermore, in the course of this project an investigation shall be conducted to evaluate which currently available ion source technology is most suited to provide mixed carbon/helium beams in the required intensity range, compatible with the MedAustron linear accelerator transport capabilities. This might include ion source tests in other installations and partner facilities.

Although the mixed beam could use the default carbon-ion optics in the MEBT, the synchrotron, and the HEBT, additional retuning would be required in some sections of the accelerator, especially the multi-turn injection which should be optimised for carbon ions, yet must provide the appropriate rate of helium-ion transport, and the mixed particle extraction from the main ring. Depending on the chosen ion source and gas type, there might be traces of other parasitic ions (e.g. oxygen) in the mixed beam. Due to the slightly different masses of oxygen, carbon, and helium, the ion beams can be adjusted on separated orbits in the synchrotron. Figure 3.1 shows on the left side the magnetic beam rigidity of $^4He^{2+}$ ions, $^{12}C^{6+}$ ions, and $^{16}O^{8+}$ ions as a function of the kinetic energy per unit mass for the clinically used energy range. Due to the magnification in the right diagram of Figure 3.1, the small differences in the magnetic beam rigidity for the three ion species can be seen, i.e. for the same kinetic energy per unit mass, helium ions have the highest and oxygen ions have the lowest magnetic beam rigidity.

A low emittance ion beam with a very low momentum spread might provide the possibility to separate parasitic beam sufficiently in certain sections of the synchrotron to dump them

on interceptive devices. This procedure could be executed during multi-turn injection, on flat bottom or after beam acceleration. On the other hand, before extraction, the remaining helium/carbon beam shall be mixed in momentum space via a respective increase of momentum spread and consecutive slow extraction, or by exciting different energy bands for different ion species using selective RFKO excitation frequencies. Beam separation in the final sector of the MEBT and within the synchrotron, as discussed above, shall be studied using particle tracking simulations. These results will determine which potential parasitic beam suppression methods can be used.

Beyond this research period, a proper ion source shall be installed in MedAustron's fourth ion source (S4) branch. Exploiting the synergy of beam commissioning efforts for $^{4}He^{2+}$ (LEBT, LINAC) and $^{12}C^{4+}$ beams (MEBT, synchrotron, HEBT), a crude mixed beam could be transported to IR 1. In a cooperation with the Radiation Physics group, the extracted beams shall be characterised and both particle ranges can be detected by means of the proton computed tomography (pCT) set-up and the available dosimetric devices for the range verification such as the Peakfinder.

WP 3.3 Low Flux Configuration

According to the request of the non-clinical user groups, several low flux settings were commissioned for protons in the clinically used energy range and also for 800 MeV in the last research period from 2019 until 2021. These dedicated beams were and are regularly used especially for the project on imaging with ion beams and for measurements with the different microdosimeters available at MedAustron. In order to extend these experiments in IR 1 and to obtain microdosimetric spectra also for carbon ions, a reduced carbon beam flux suitable for the used microdosimeters would be beneficial. With the already collected experiences for protons, the beam time demand for establishing a first carbon ion low flux setting can be estimated with about three shifts. In the course of the helium-ion commissioning, dedicated low flux settings would be very appreciated, especially for exploring helium ions for particle imaging as already described above.

Main Involved Personnel and Timeline

Almost all above mentioned topics depend strongly on the availability of diploma and project students with a technical background.

Required Beam Time

work package	2022	2023	2024	total
WP 1 – Implementation of Helium Ions	25	28	23	76
WP 2 – Extraction Mechanisms	15	14	24	53
WP 3 – Novel Developments	2	4	4	10
total	**42**	**46**	**51**	**139**

Table 3.1: Estimated beam time in shifts per year for "Accelerator Physics".

Acronyms

4DCT	four-dimensional computed tomography.
AI	artifical intelligence.
ARDOS	advanced radiation dosimetry system.
ART	adaptive radiation therapy.
ASIC	application-specific integrated circuit.
ATM	ataxia telangiectasia mutated.
ATMi	ataxia telangiectasia mutated inhibitor.
ATR	ataxia telangiectasia and Rad3-related protein.
ATRi	ataxia telangiectasia and Rad3-related protein inhibitor.
BEC	Buck Engineering and Consulting.
BPND	binding potential with non-displaceable tracer uptake.
BPP	binding potential plasma.
CBCT	cone beam computed tomography.
CERN	Conseil Européen pour la Recherche Nucléaire.
CIRT	carbon-ion radiotherapy.
CNAO	Centro Nazionale di Adroterapia Oncologica.
COSE	constant optics slow extraction.
CT	computed tomography.
DµD	dosimeter-microdosimeter.
DCE	dynamic contrast enhanced.
DDR	DNA damage response.
DNA	deoxyribonucleic acid.
DWI	diffusion-weighted.
EC	European Commission.
ECR	electron cyclotron resonance.
FFG	Austrian Research Promotion Agency.
FPGA	field programmable gate array.
FWF	Austrian Science Fund.
GAN	generative adversarial network.

GATE	Geant4 application for tomographic emission.
HEBT	high-energy beam transport.
HEPHY	Institute of High Energy Physics.
HiBPM	Hi-Precision Beam Position and Intensity Monitor.
HIT	Heidelberg Ion-Beam Therapy Centre.
HITRI*plus*	Heavy Ion Therapy Research Integration *plus*.
HNSCC	head and neck squamous cell carcinoma.
HR	homologous recombination.
HSP90i	heat-shock-protein 90 inhibitor.
HU	Hounsfield unit.
IAEA	International Atomic Energy Agency.
IBA	Ion Beam Applications.
ICRU	International Commission on Radiation Units and Measurements.
IDEAL	independent dose calculation.
IMPACTOR	Image Guided Particle Therapy of Uveal Melanoma with a Multiport Beam Arrangement.
INFN	Istituto Nazionale di Fisica Nucleare.
IR 1	irradiation room 1.
IR 2	irradiation room 2.
ITN	Intensive Training Network.
LEBT	low-energy beam transport.
LET	linear energy transfer.
LGAD	low gain avalanche detector.
LINAC	linear accelerator.
MAPTA	MedAustron particle therapy accelerator.
MATEO	MedAustron treatment environment for ocular tumours.
MBH	horizontal bending (dipol) magnet.
MC	Monte Carlo.
MEBT	medium-energy beam transport.
MEE	multi-energy extraction.
MIP	minimum ionising particle.
MKM	microdosimetric kinetic model.
mpMRI	multi-parametric magnetic resonance imaging.
MR	magnetic resonance.
MRF	magnetic resonance fingerprinting.
MRgPT	magnetic resonance guided particle therapy.
MRI	magnetic resonance imaging.
MRO2	oxygen metabolic rate.
MRT	magnetic resonance tomography.
NEMA	National Electrical Manufactures Association.
NPL	National Physics Laboratory.
NSCLC	non-small cell lung carcinoma.

OAR	organ at risk.
OEF	oxygen extraction fraction.
OncoRay	Centre for Radiation Research in Oncology.
OS	osteosarcoma.
PAIR	Pre-clinicAl Ion beam Research.
PARP	poly ADP-ribose polimerase.
PARPi	poly ADP-ribose polymerase inhibitor.
PBS	pencil-beam scanning.
PC	pancreatic cancer.
pCT	proton computed tomography.
PDAC	pancreatic ductal adenocarcinoma.
PET	positron emission tomography.
PExTex	Planetary Exploration Textiles.
PSQA	patient-specific quality assurance.
PSS	phase space stepper.
QA	quality assurance.
qBOLD	quantitative blood oxygenation level dependent.
qPCR	quantitative polymerase chain reaction.
RAPTOR	Real Time Adaptive Particle Therapy of Cancer.
RBE	relative biological effectiveness.
RF	radio frequency.
RFKO	radio frequency knock-out.
RFQ	radio-frequency quadrupole.
S3	third ion source.
S4	fourth ion source.
sCT	synthetic computed tomography.
SDR	software defined radio.
SiC	silicon carbide.
SOBP	spread-out Bragg peak.
SOP	standard operating procedure.
SPECT	single-photon emission computed tomography.
SRT	stereotactic radiation therapy.
SVD	Silicon Vertex Detector.
TDC	time-to-digital converter.
TEPC	tissue equivalent proportional counter.
TIGRE	tomographic iterative GPU-based reconstruction toolbox.
TLD	thermoluminescent dosimeter.
TME	tumour microenvironment.
TNBC	triple-negative breast cancer.
TOF	time-of-flight.
TOLD	tissue oxygen level dependent.

TPS	treatment planning system.
TU	Technische Universität.
UHF	ultra high-field.
UMCG	University Medical Centre Gronigen.
VAM	vascular architectural mapping.